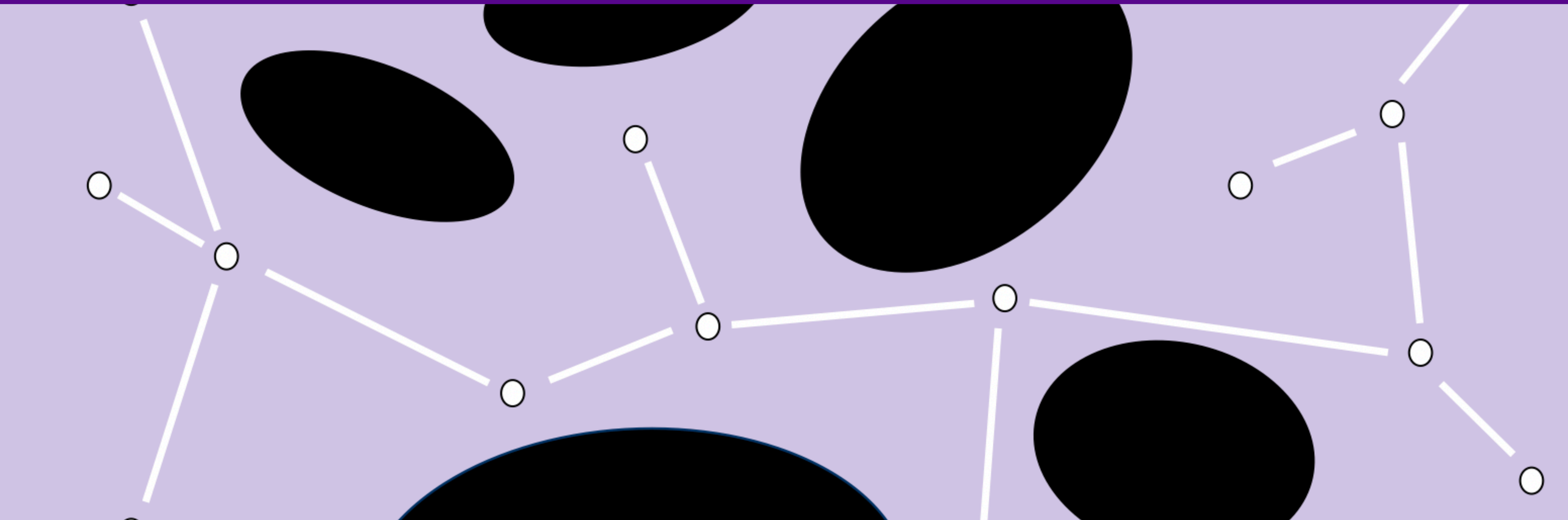
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Robot Motion & Planning

Lecture 11A - Probabilistic Road Maps



PRMs - Probabilistic Road Maps

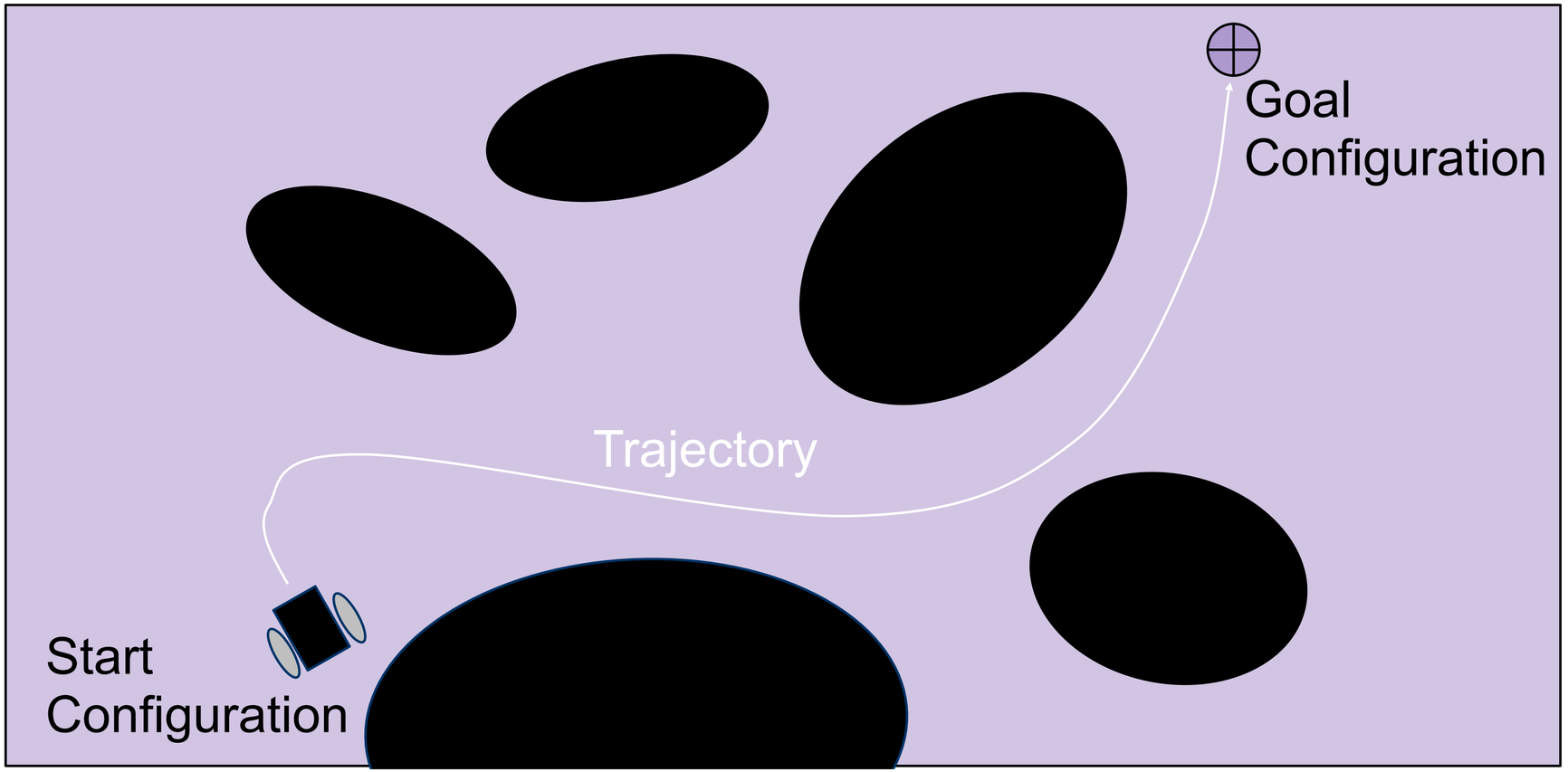
1. **Single Query**
   1. Build a new tree to search for every query
   2. Variants of RRTs - Rapidly exploring Random Trees
2. **Multi-Query**
   1. Build a single tree to search for many queries

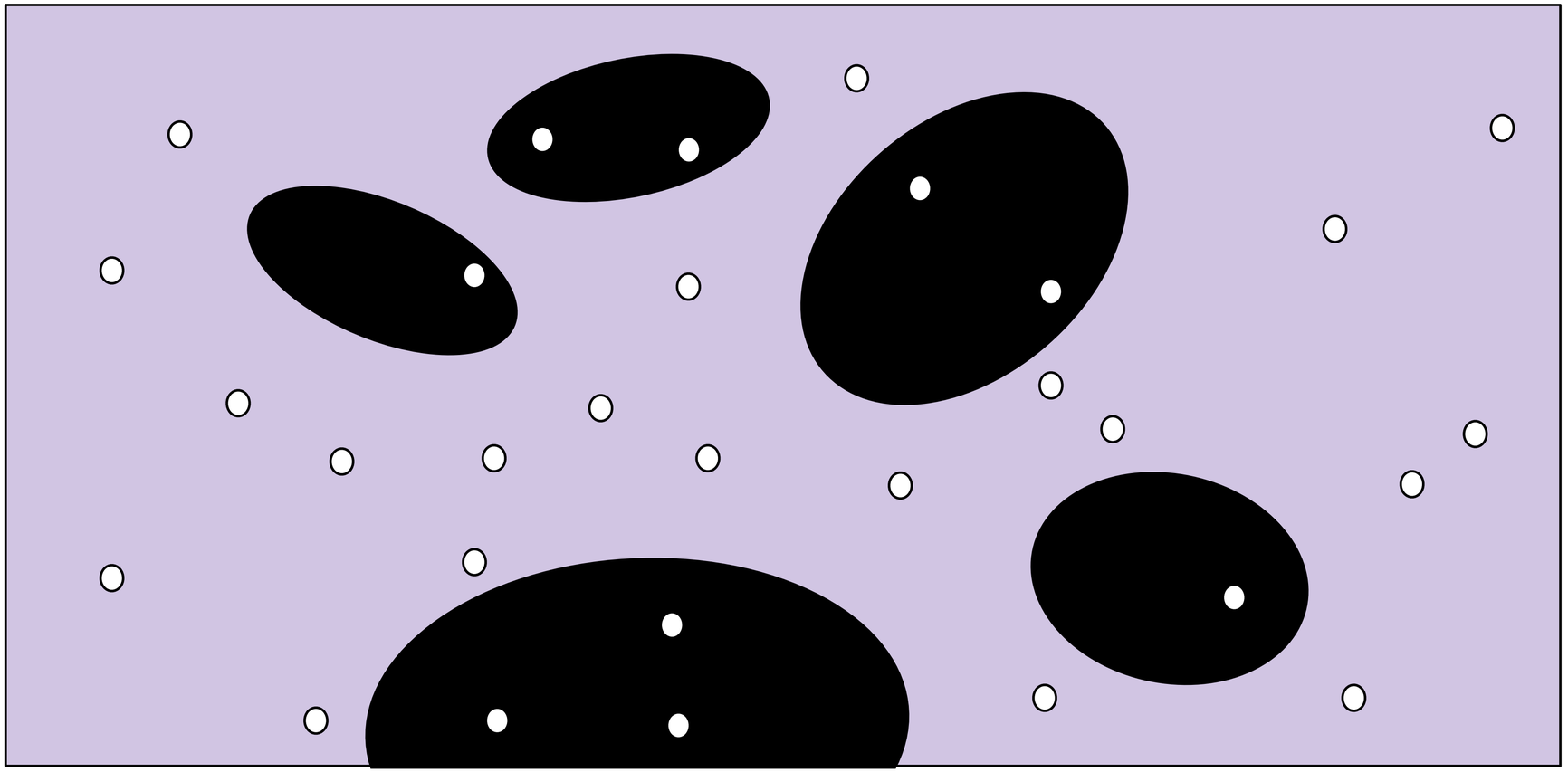
Agenda

1. Algorithm run-through
2. Pseudo Code
3. Example Implementation

Agenda

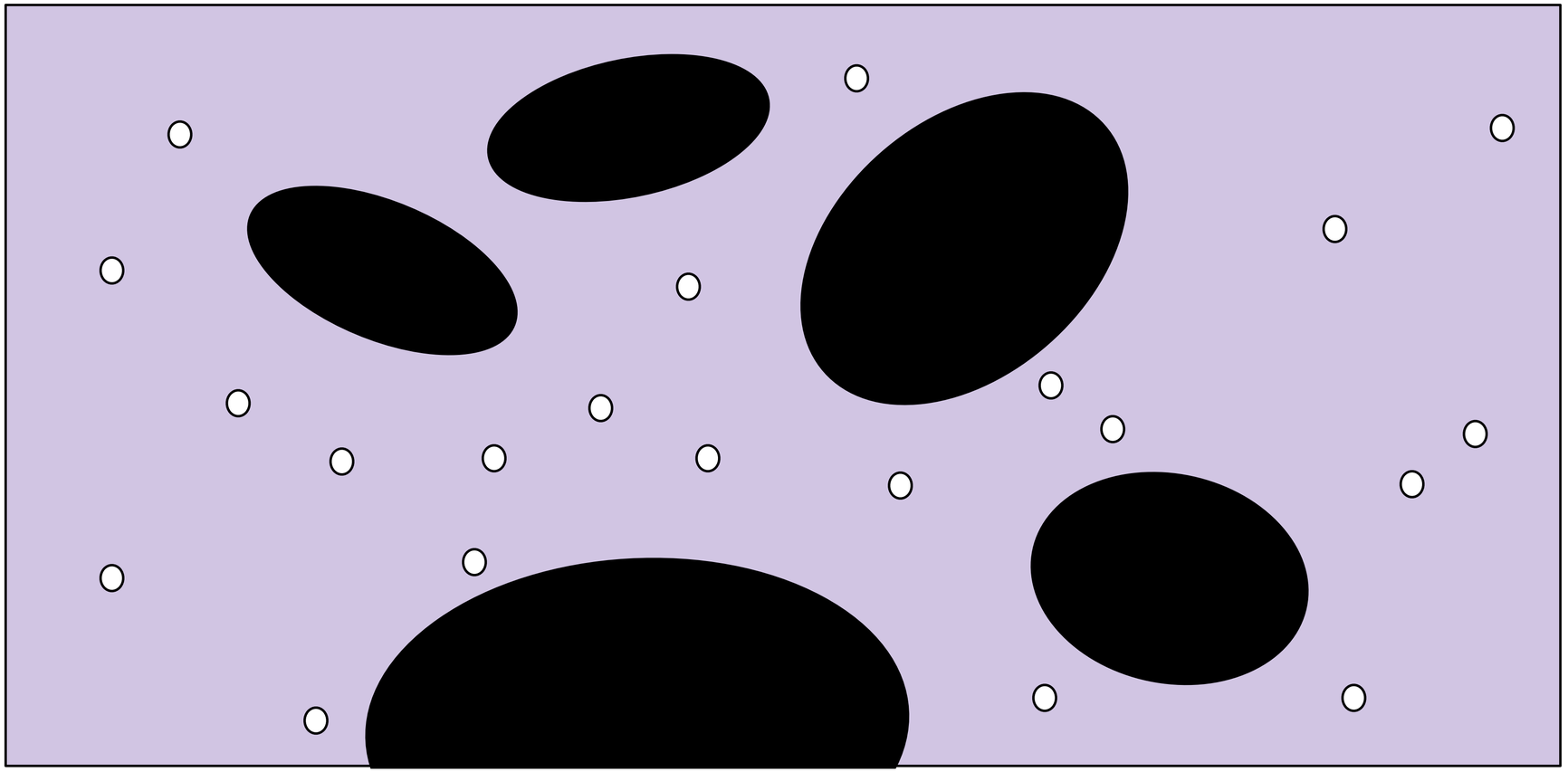
1. **Algorithm run-through**
2. Pseudo Code
3. Example Implementation


Goal
Configuration
Trajectory
Start
Configuration



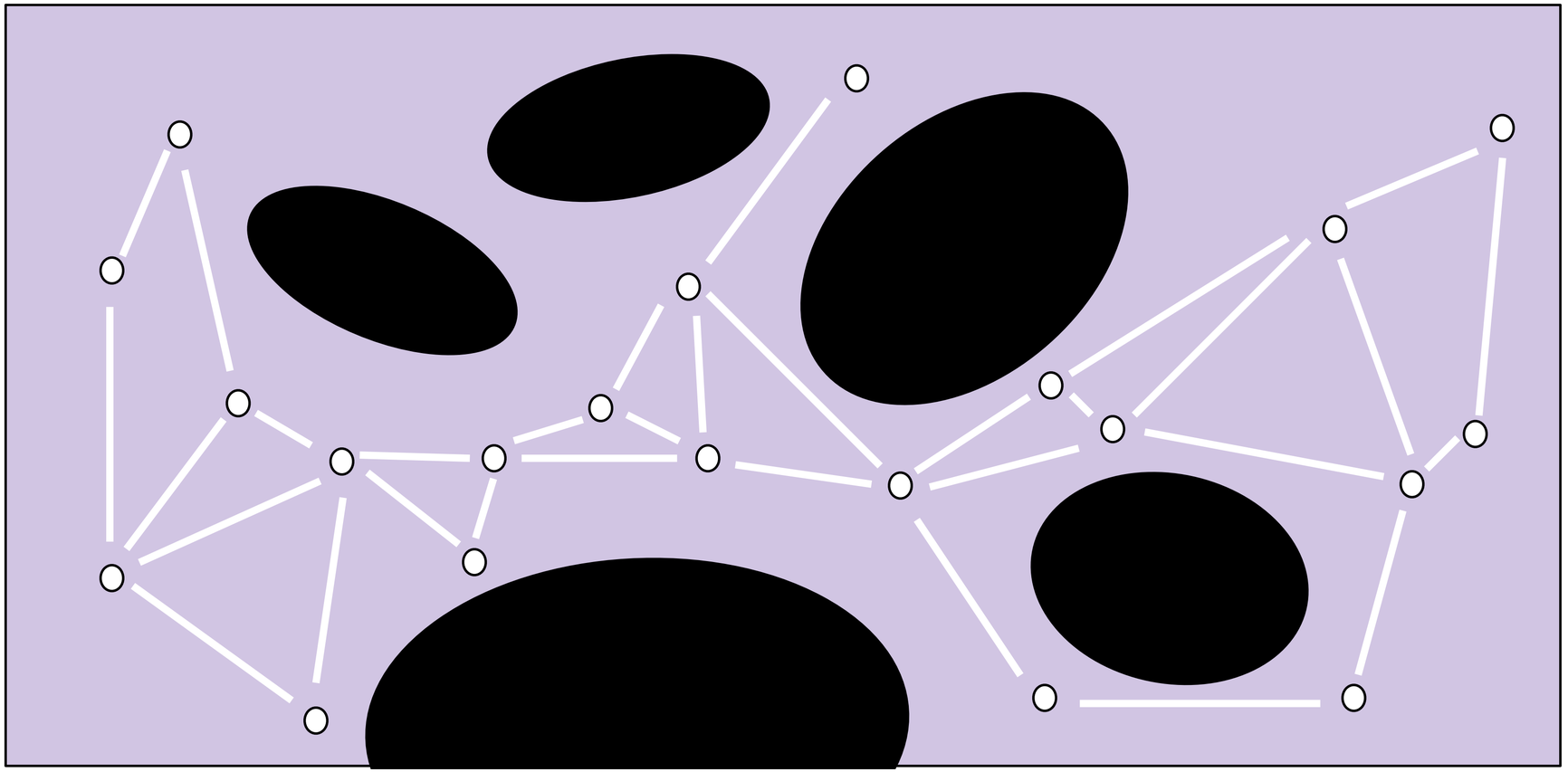
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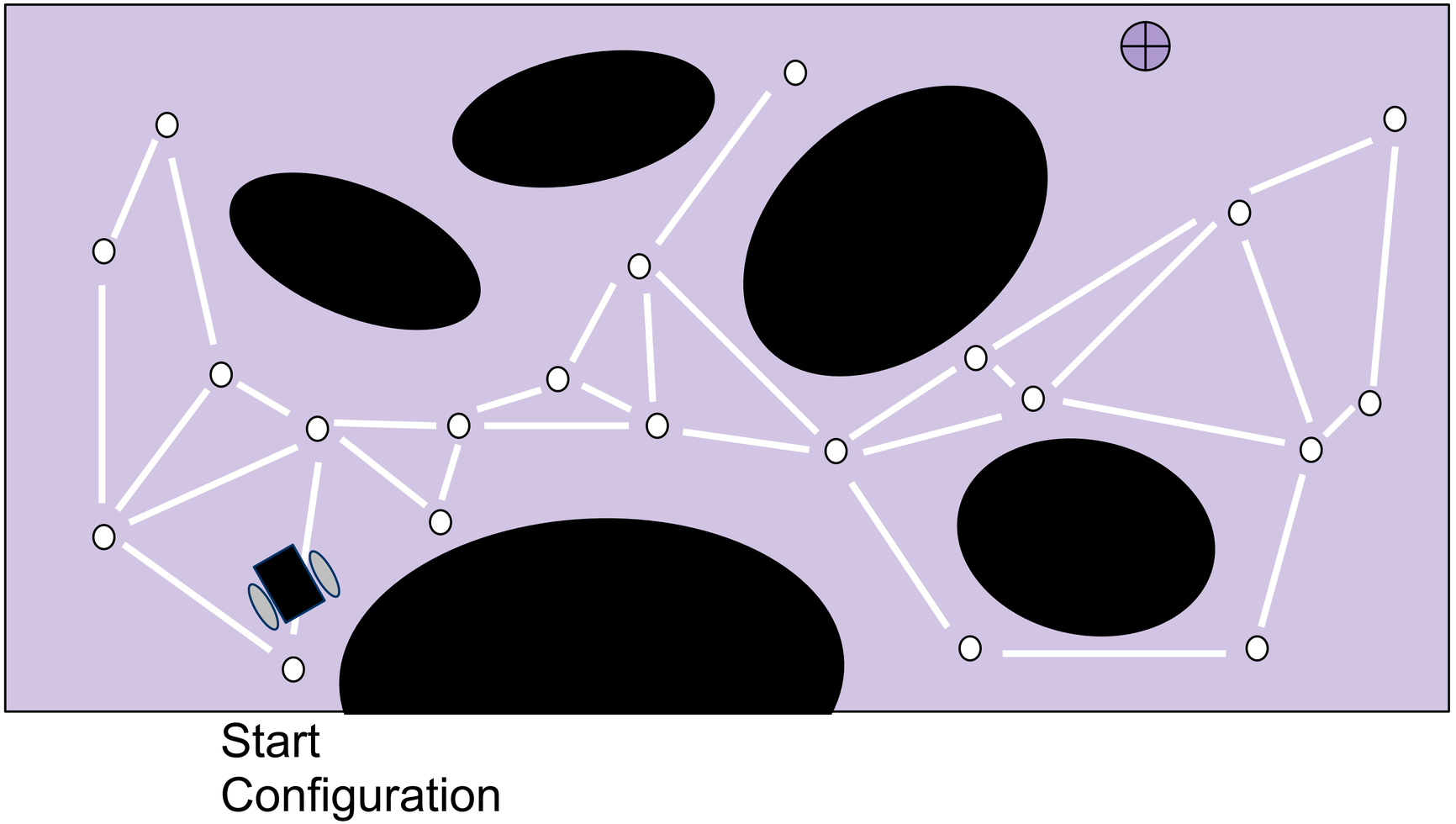
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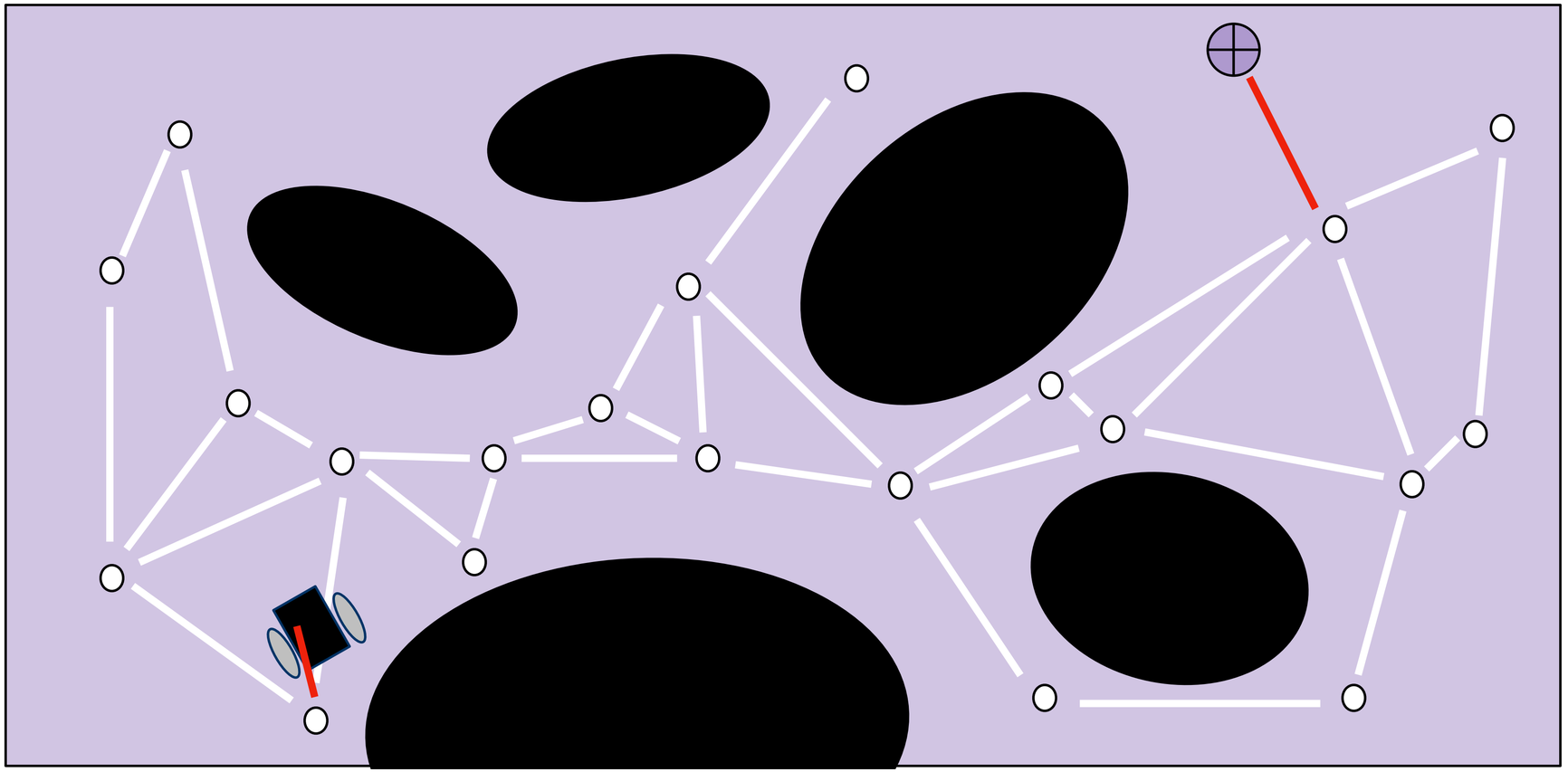


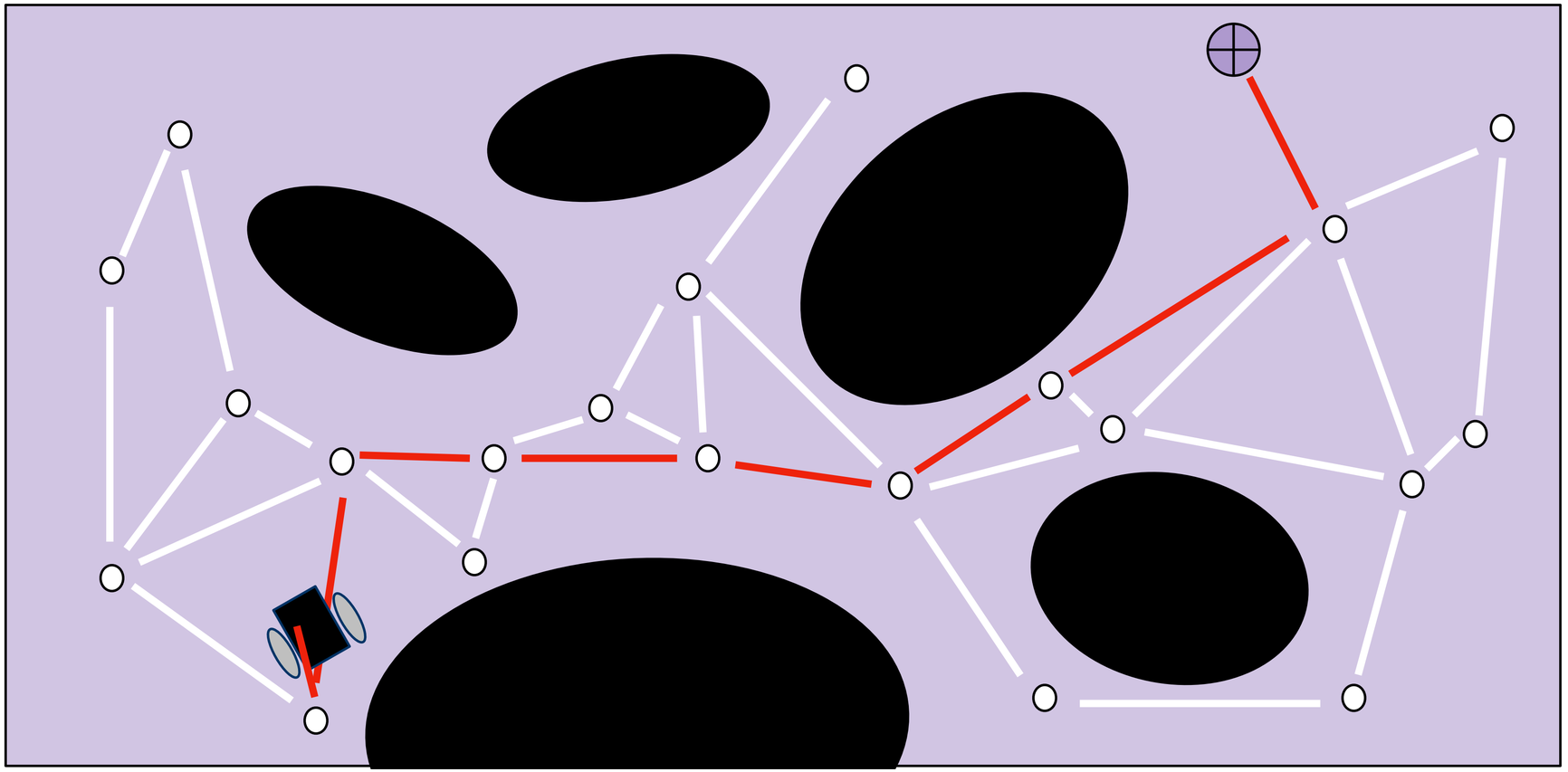
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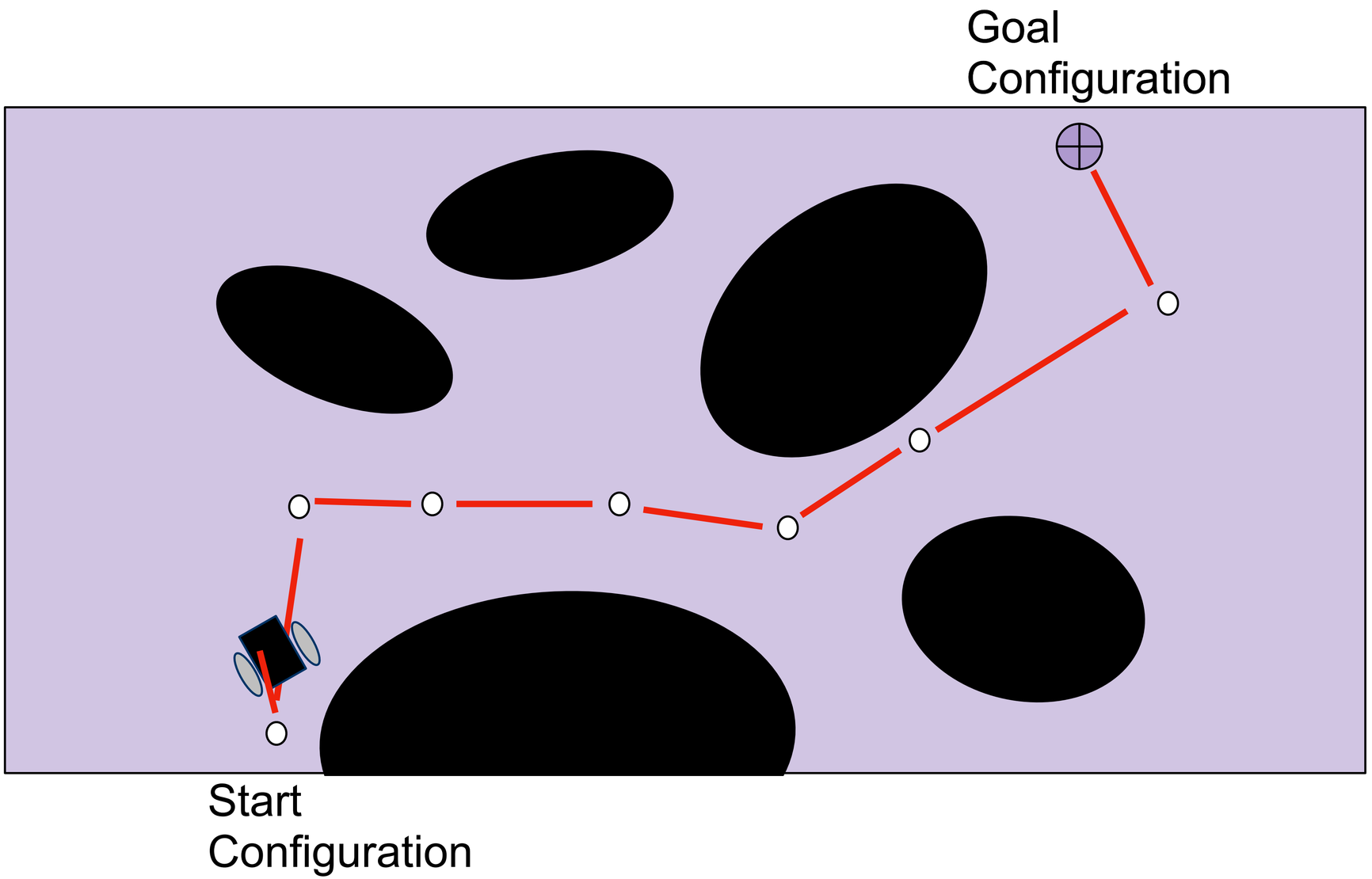
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Goal
Configuration
Start
Configuration


Goal
Configuration
Start
Configuration


Goal
Configuration
Start
Configuration


Goal
Configuration
Start
Configuration

Agenda

1. Algorithm run-through
2. **Pseudo Code**
3. Example Implementations

Multi-Query Strategy

1. Learning Phase

- Generate the PRM with two steps:
  1. Construction
  2. Expansion

2. Query Phase

- Connect start and goal configurations to PRM
- Perform a graph search to find path

Multi-Query Strategy

Nomenclature

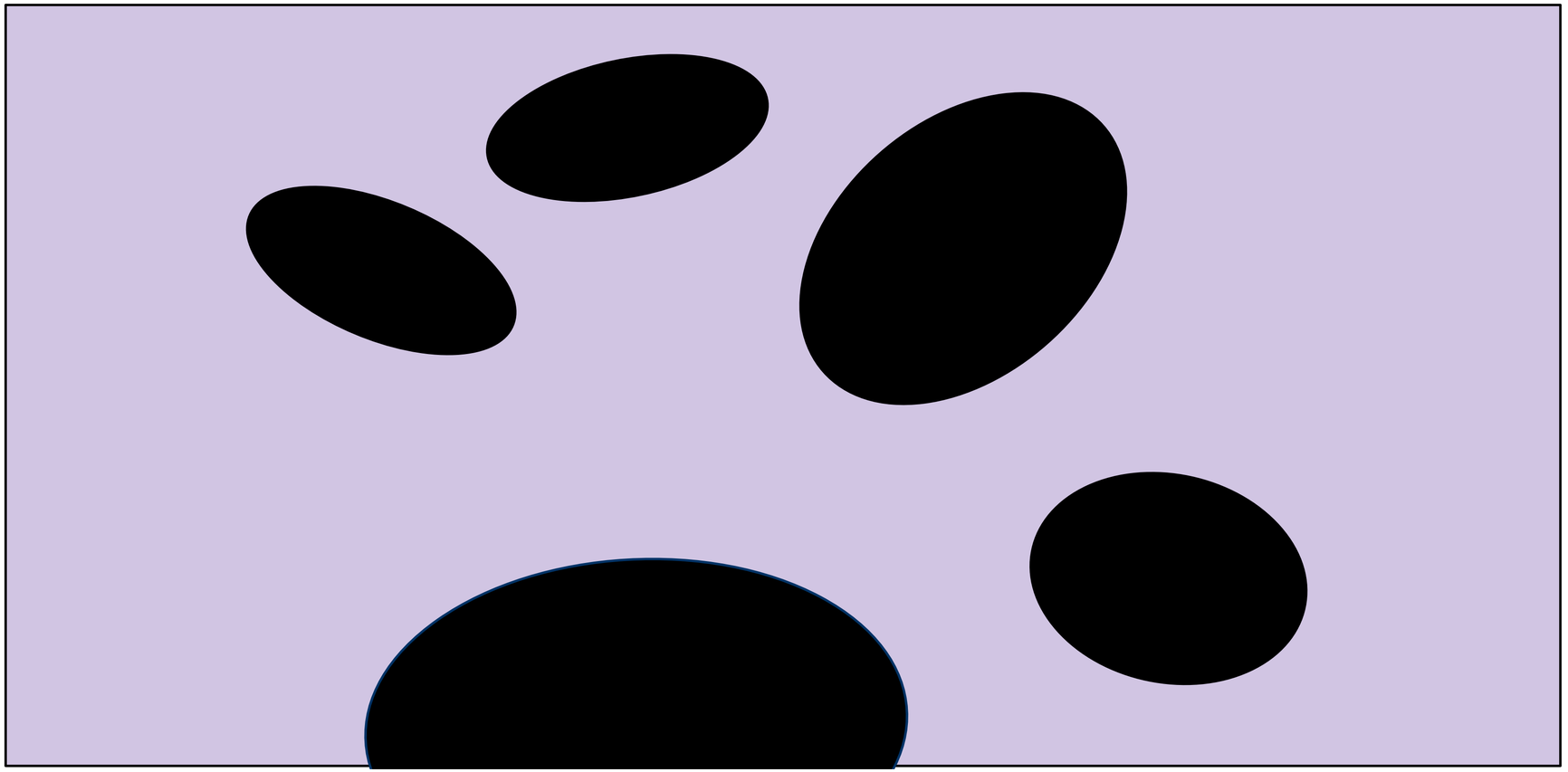
$R(N,E)$	RoadMap
N	Set of Nodes
E	Set of edges
c	Configuration
e	edge

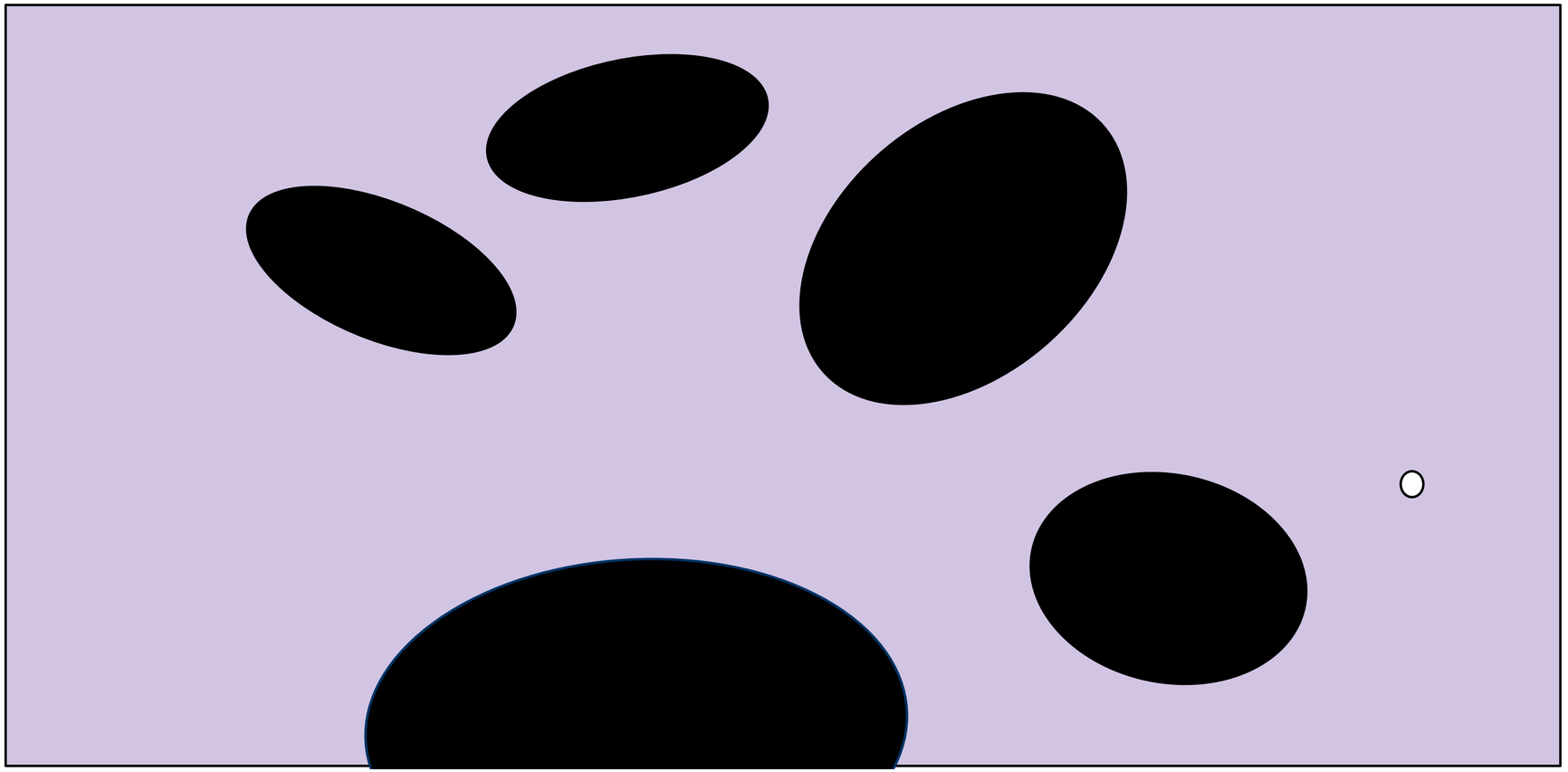
Learning Phase

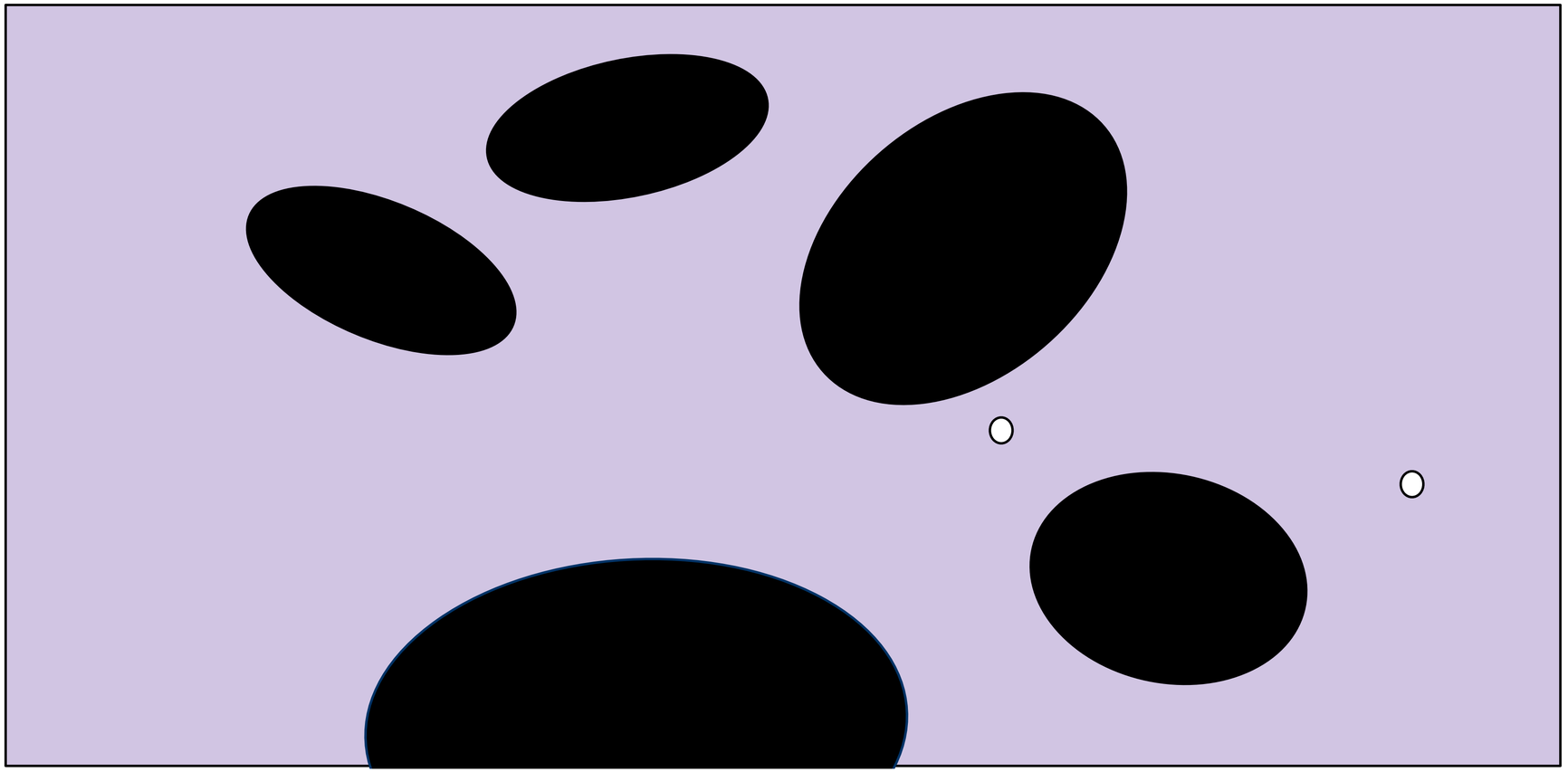
1. Construction Step

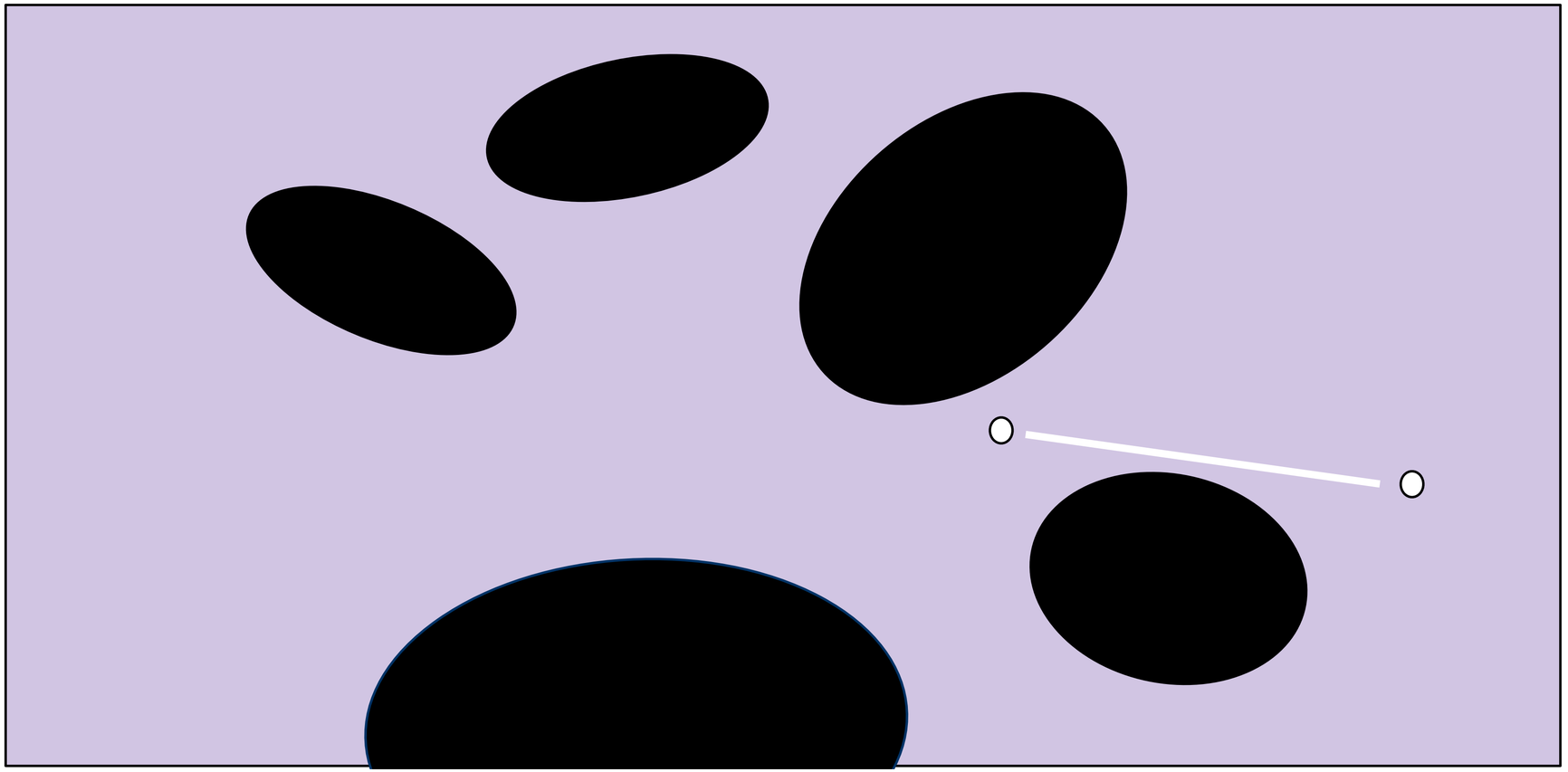
$R(N,E) = \{\}$
while (not done)

Generate a random free config c and add to N
Choose a subset N_c of candidate neighbors around c from N
Find edge e that connects c to closest node in N_c with local planner
Add e to E



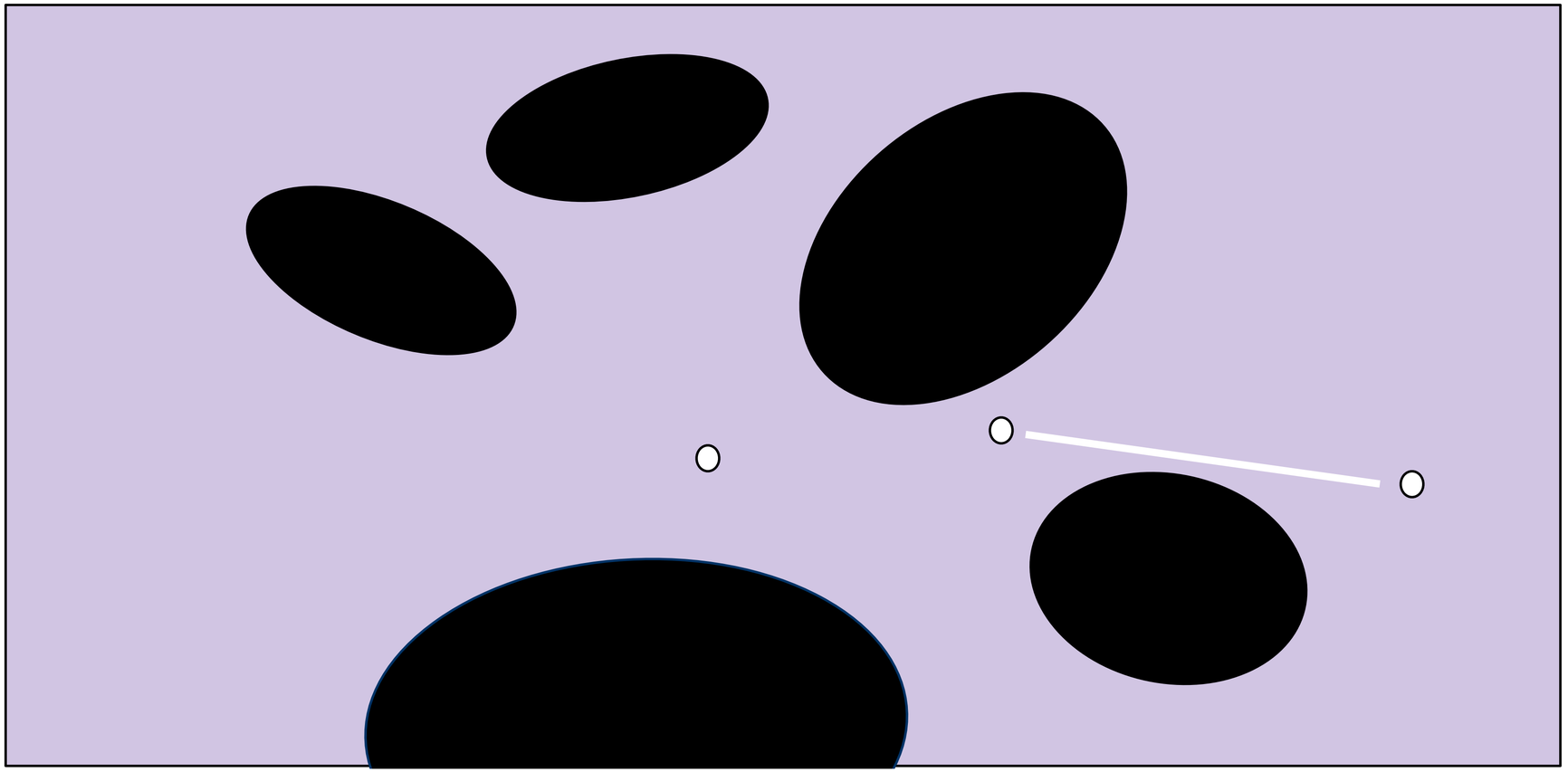






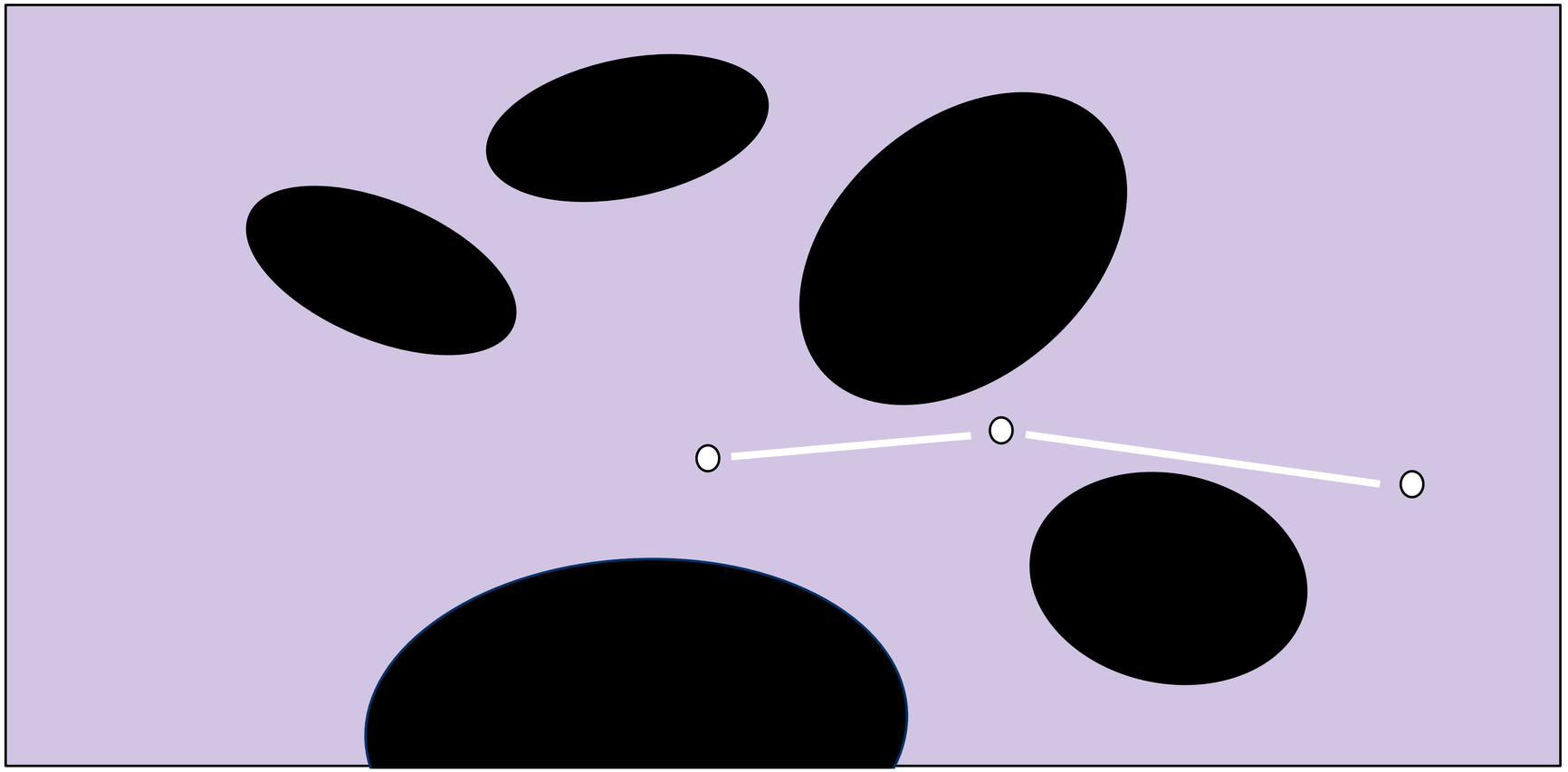
ROB-UY 3303

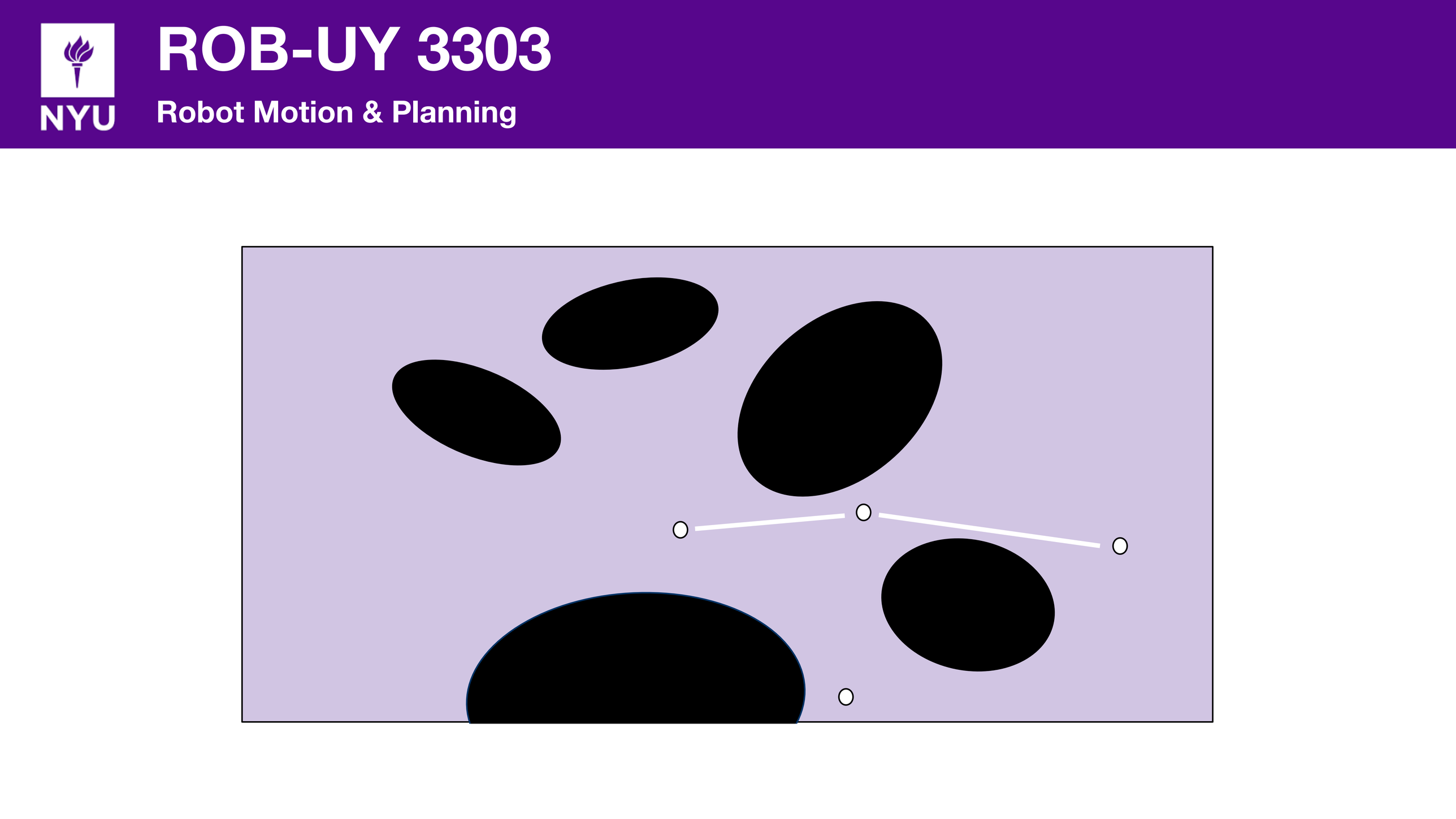
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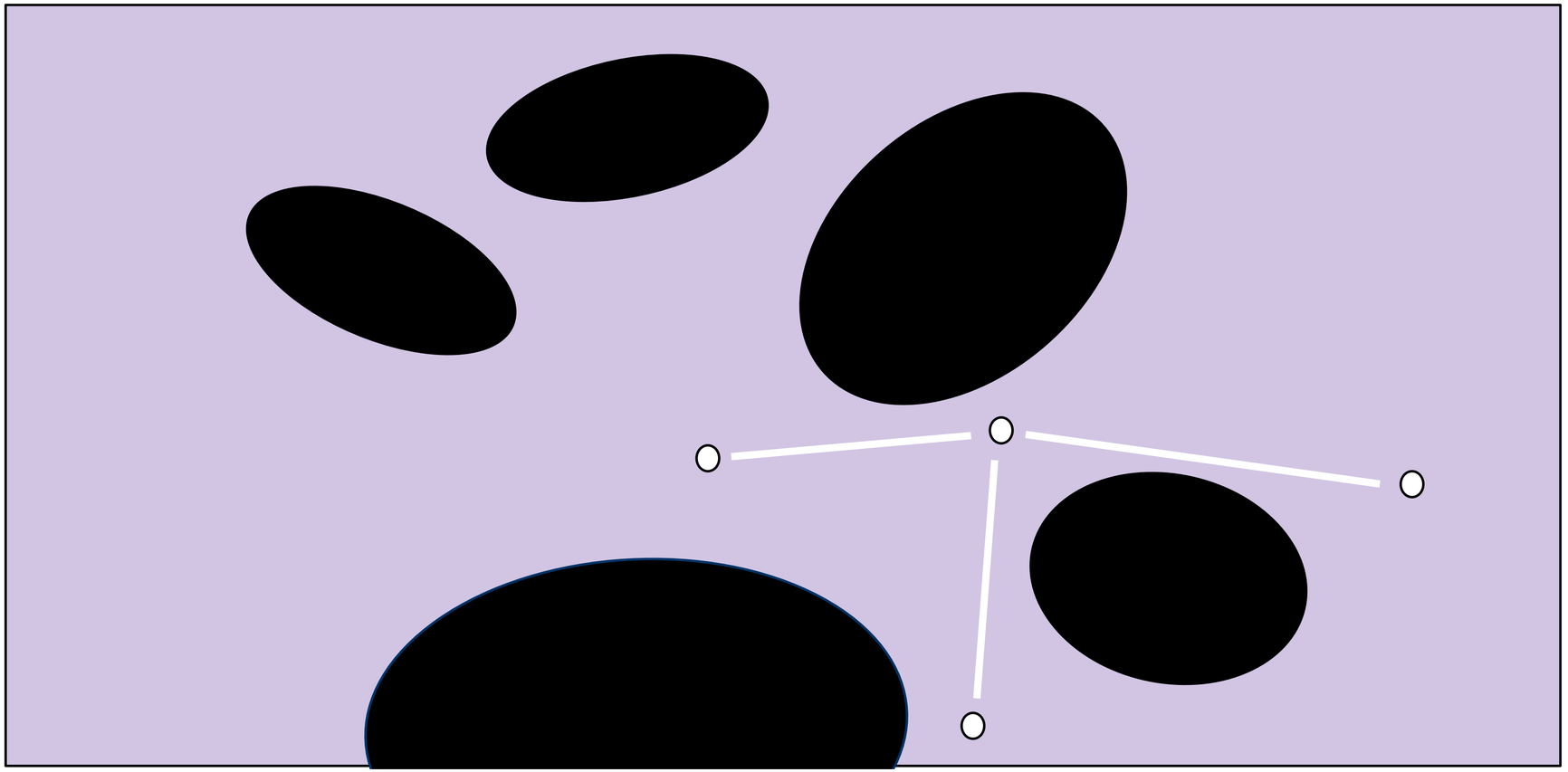


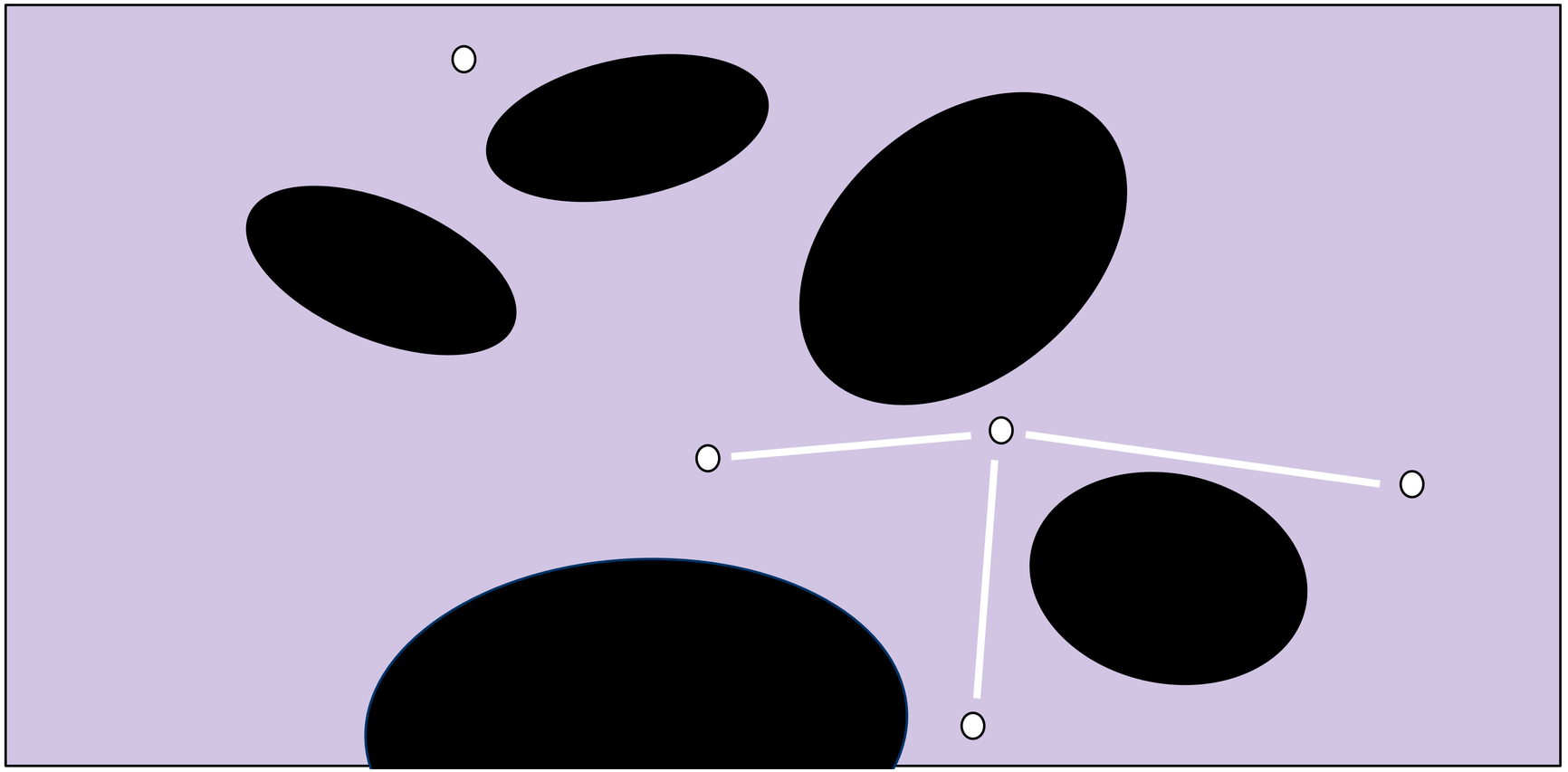
ROB-UY 3303

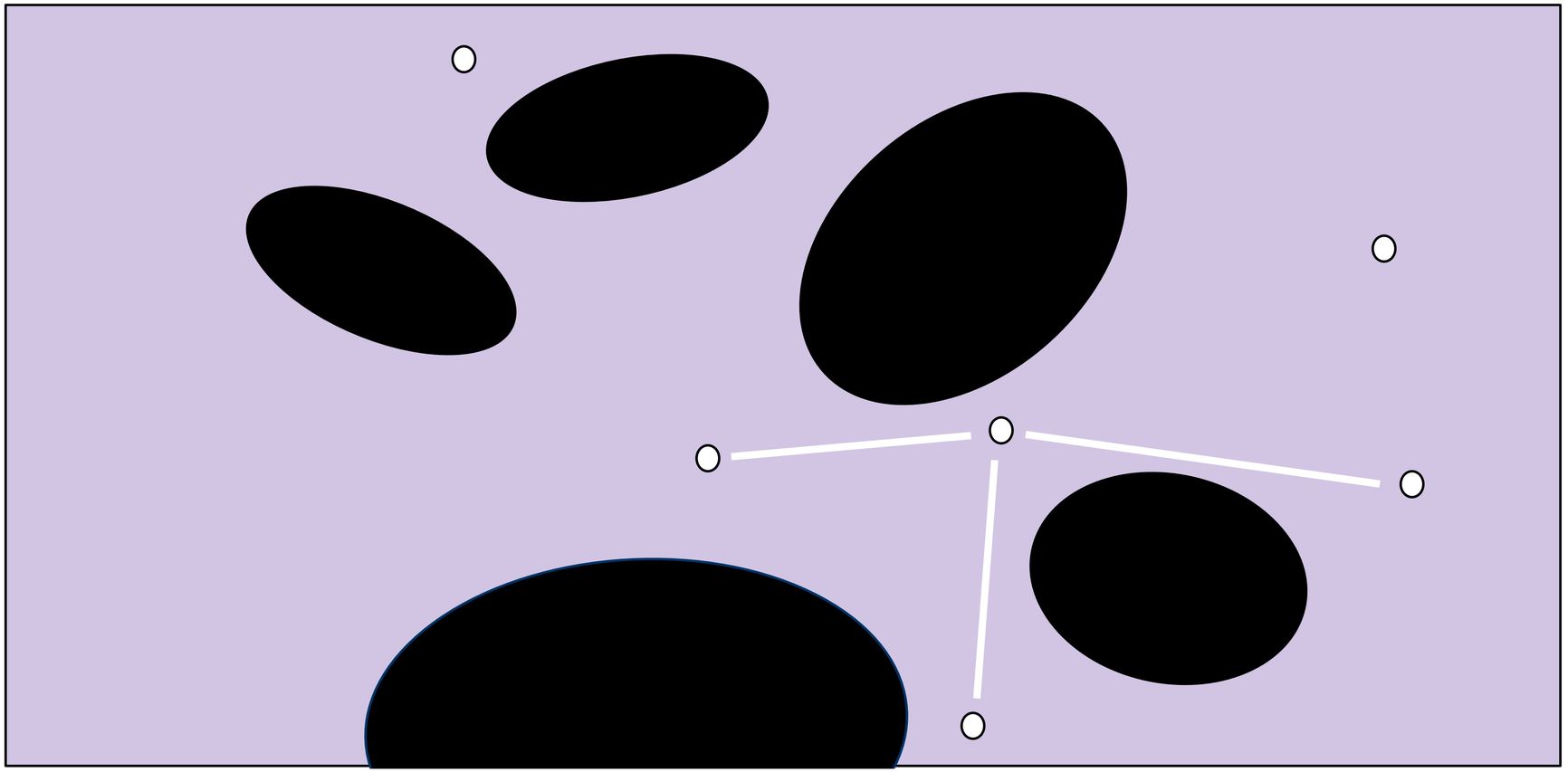
Robot Motion & Planning

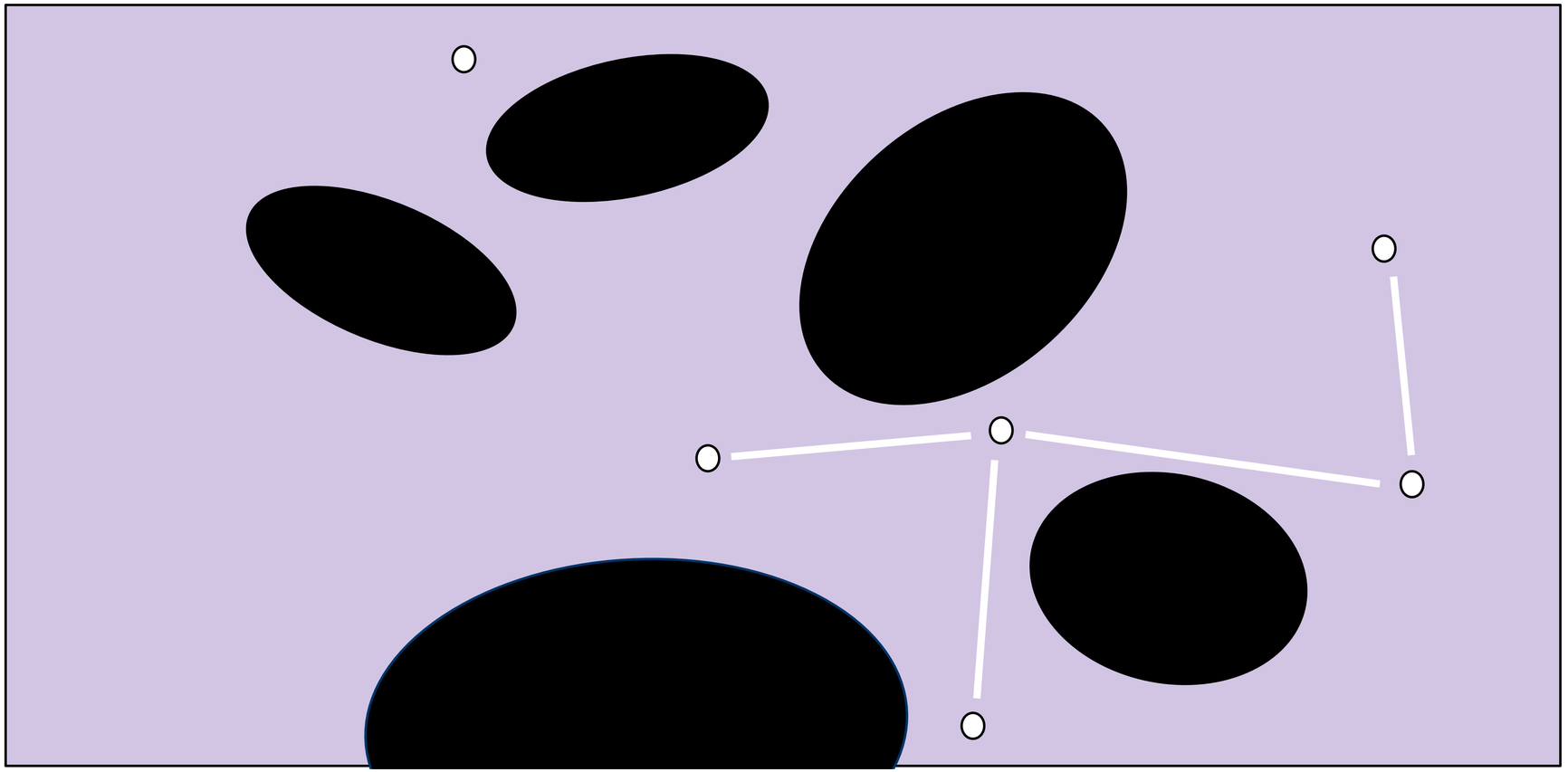



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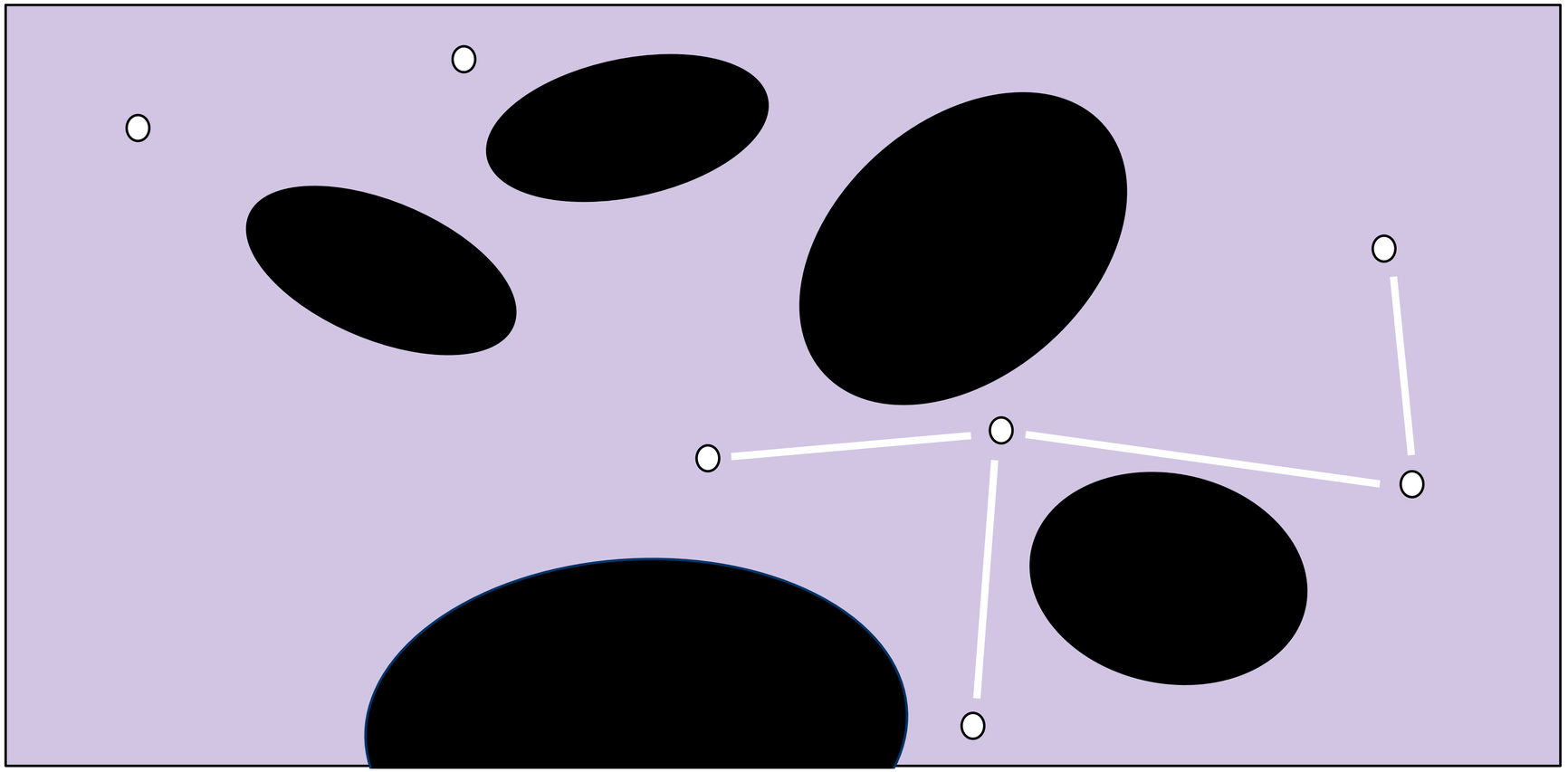






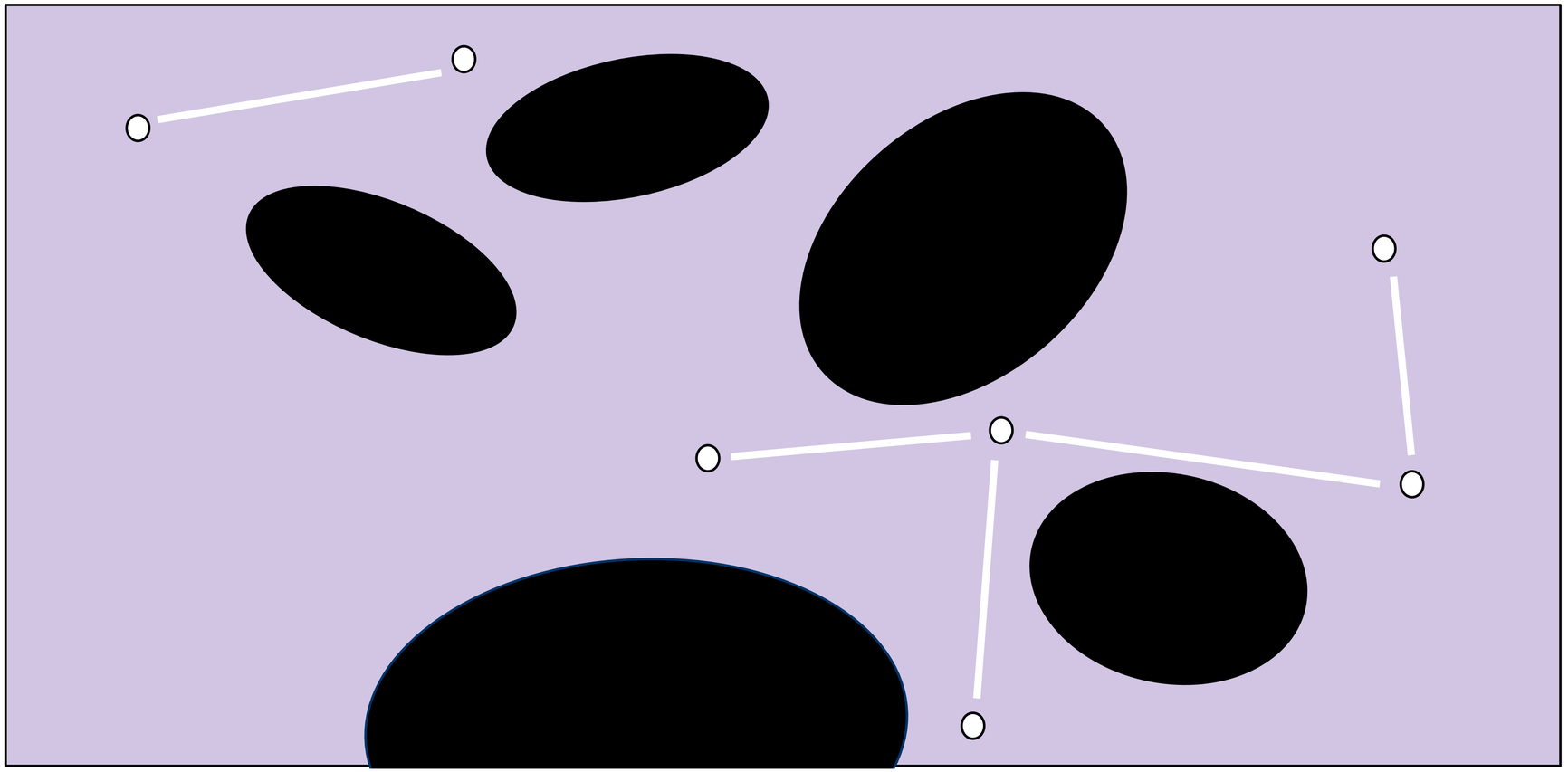
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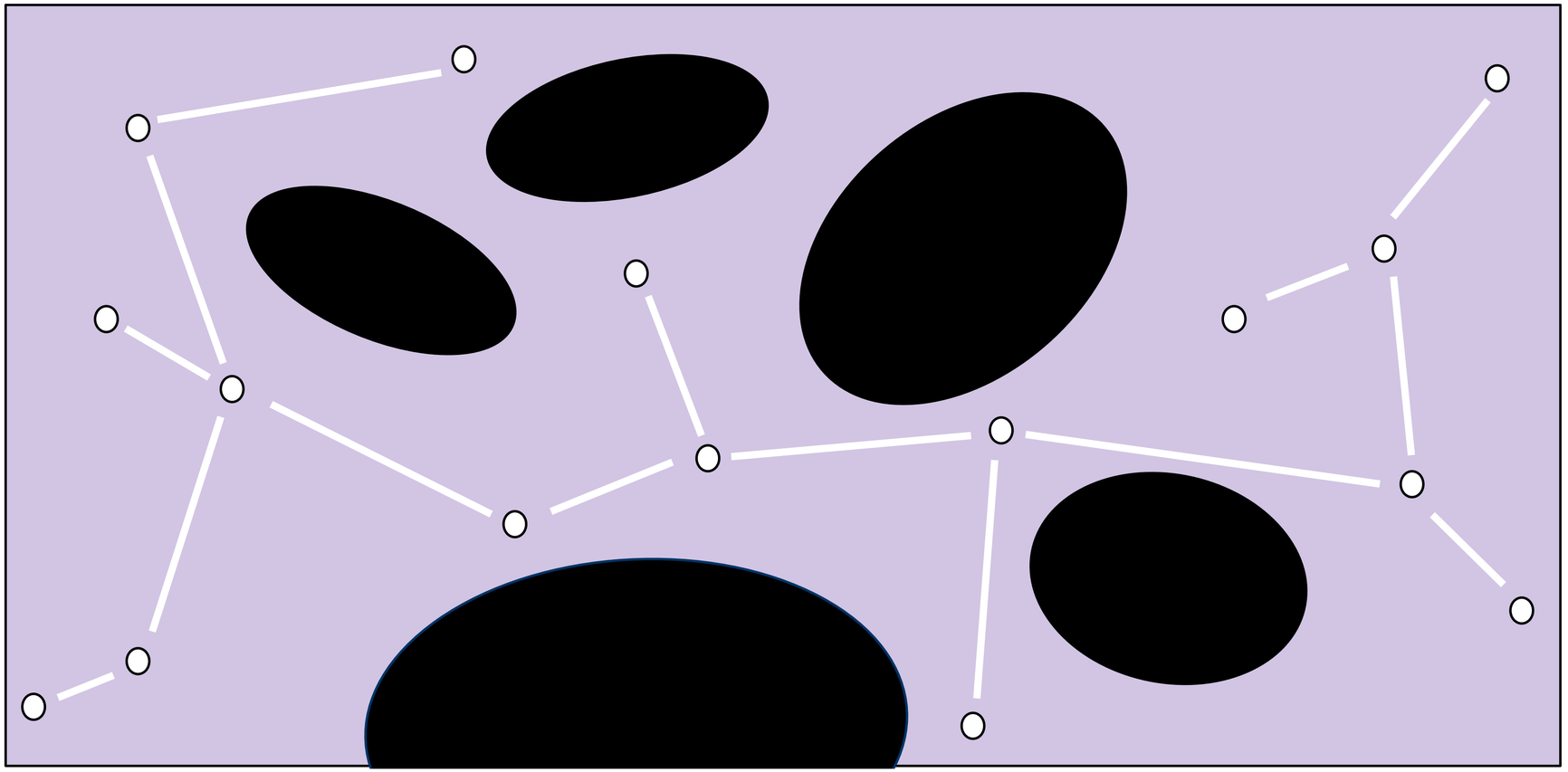
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Learning Phase

1. Construction Step - Local Planner

- Used to connect two nodes
- Must contain collision-check
- For good performance, the LP must be:
  1. Deterministic - Eliminates the need for storing local plans.
  2. Fast - To ensure quick planning queries.

Learning Phase

2. Expansion Step

Find the nodes in ‘difficult’ regions using heuristic weight function $w(c)$
Expand c using random-bounce walks
Repeat as necessary

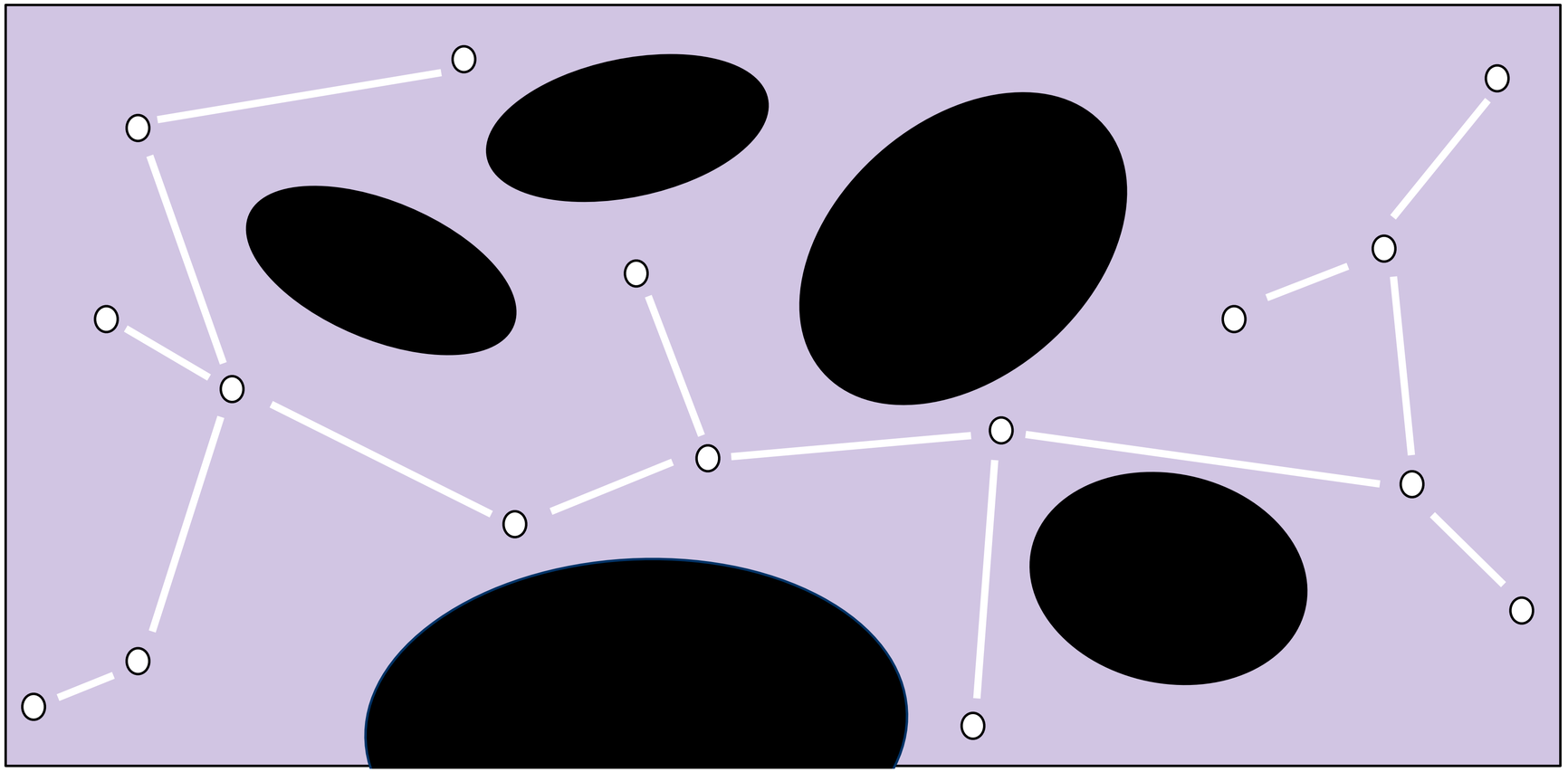
Learning Phase

2. Expansion Step

- Options for weight function $w(c)$
  - Inversely proportional to the "number of nodes within some predefined distance from c
  - Inversely proportional to the "distance from c to the nearest connected component not containing c
  - Proportional to the "failure ratio of the local planner"

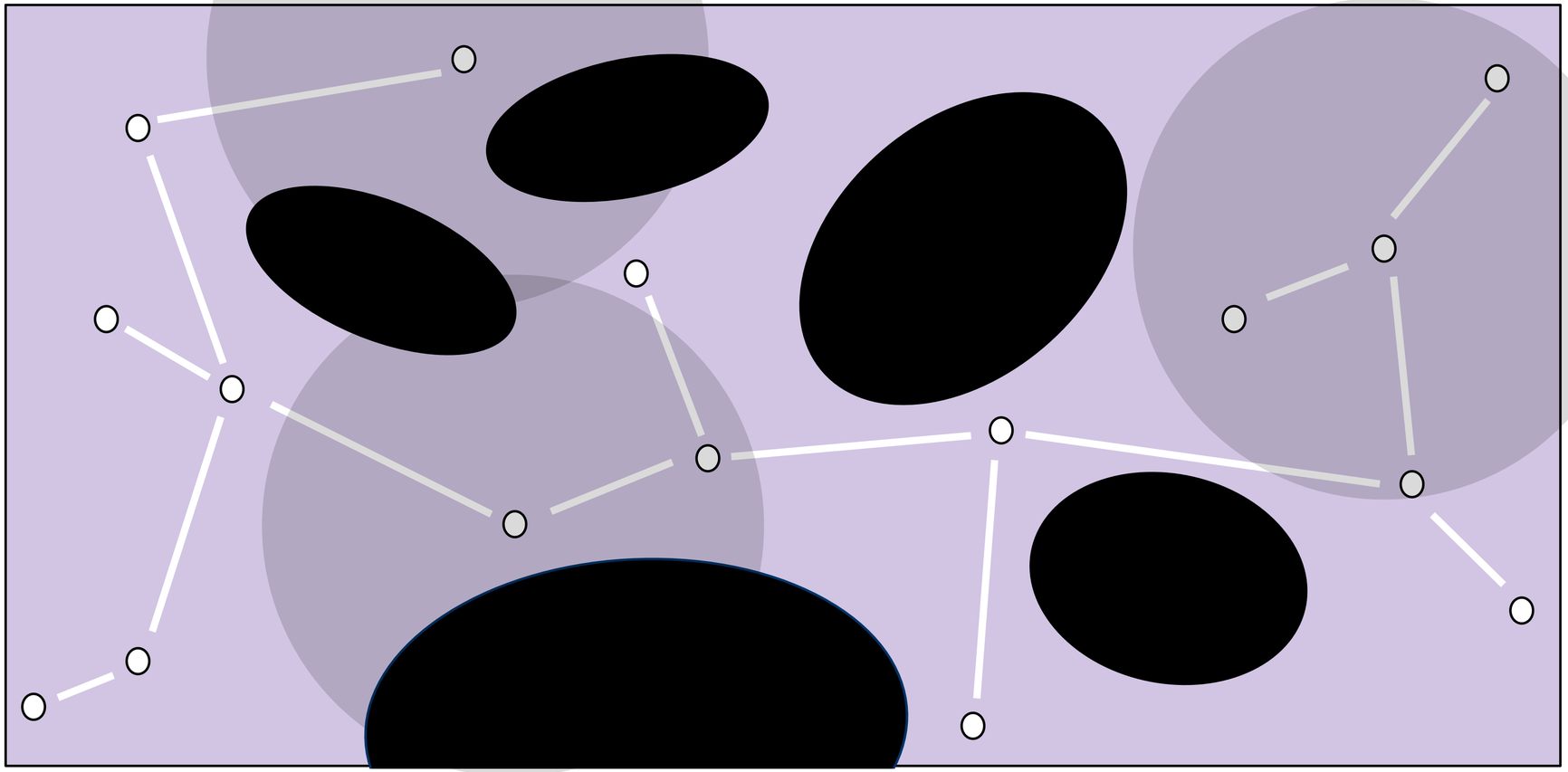
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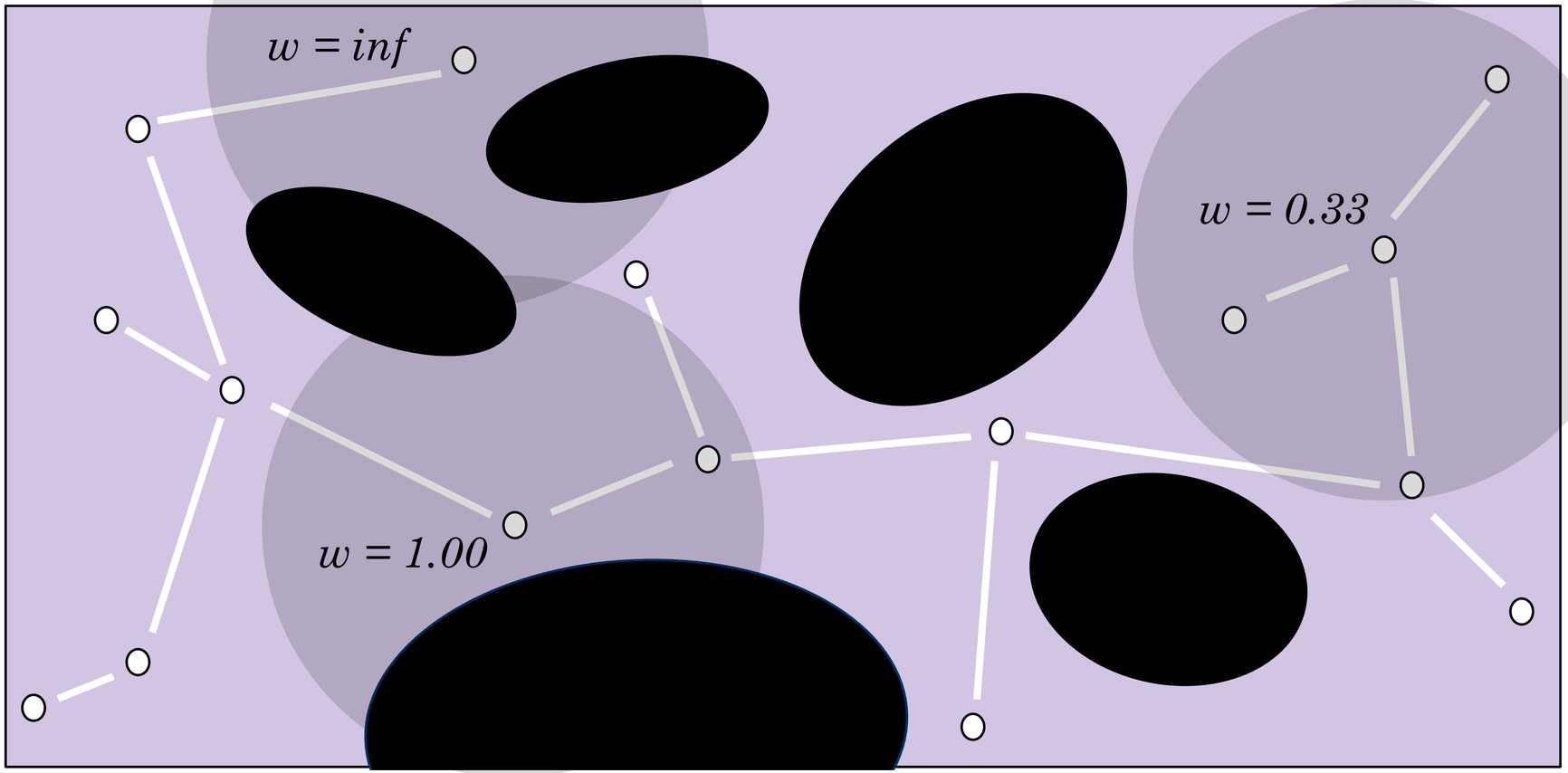
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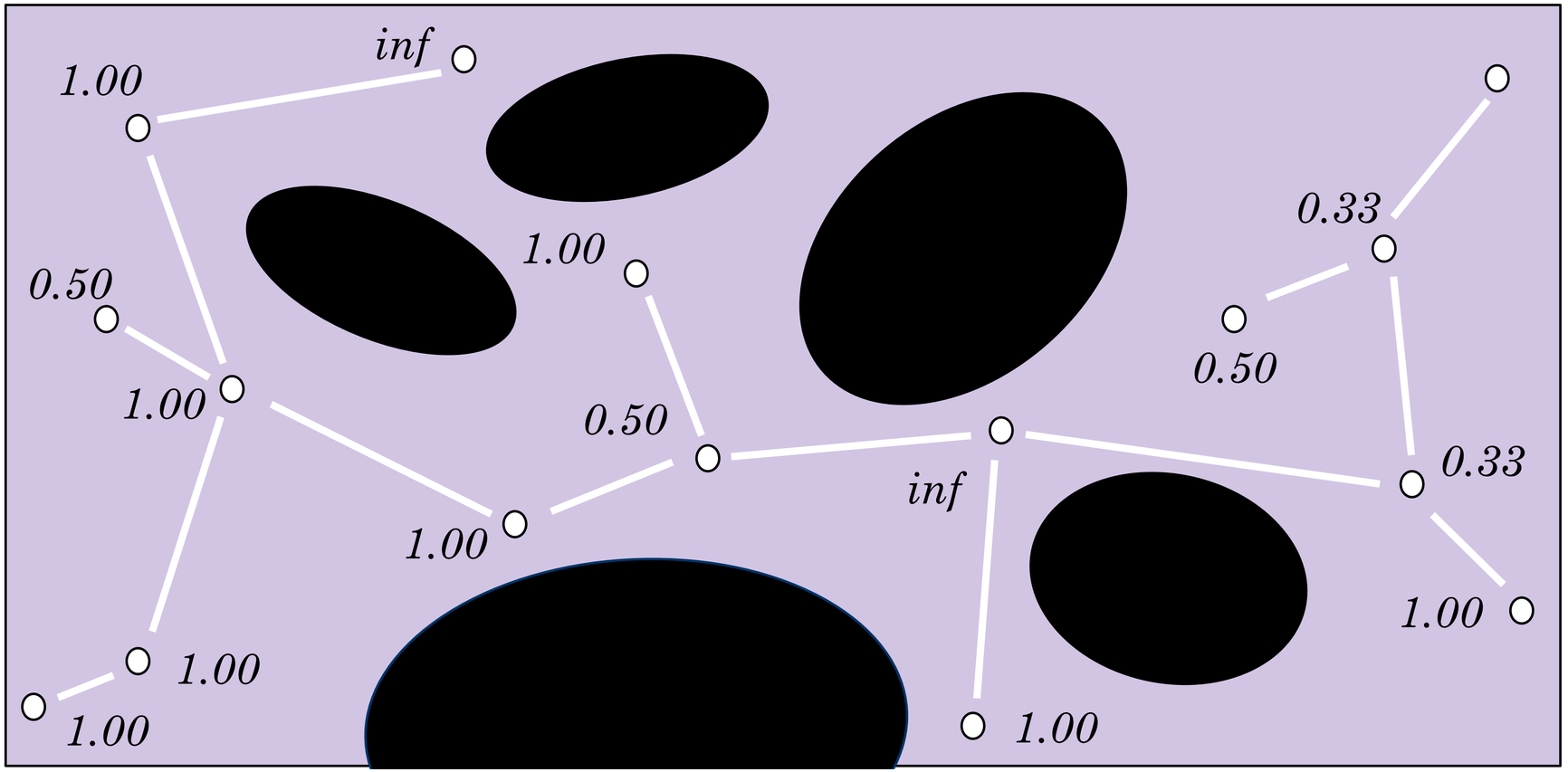


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w = inf
w = 0.33
w = 1.00

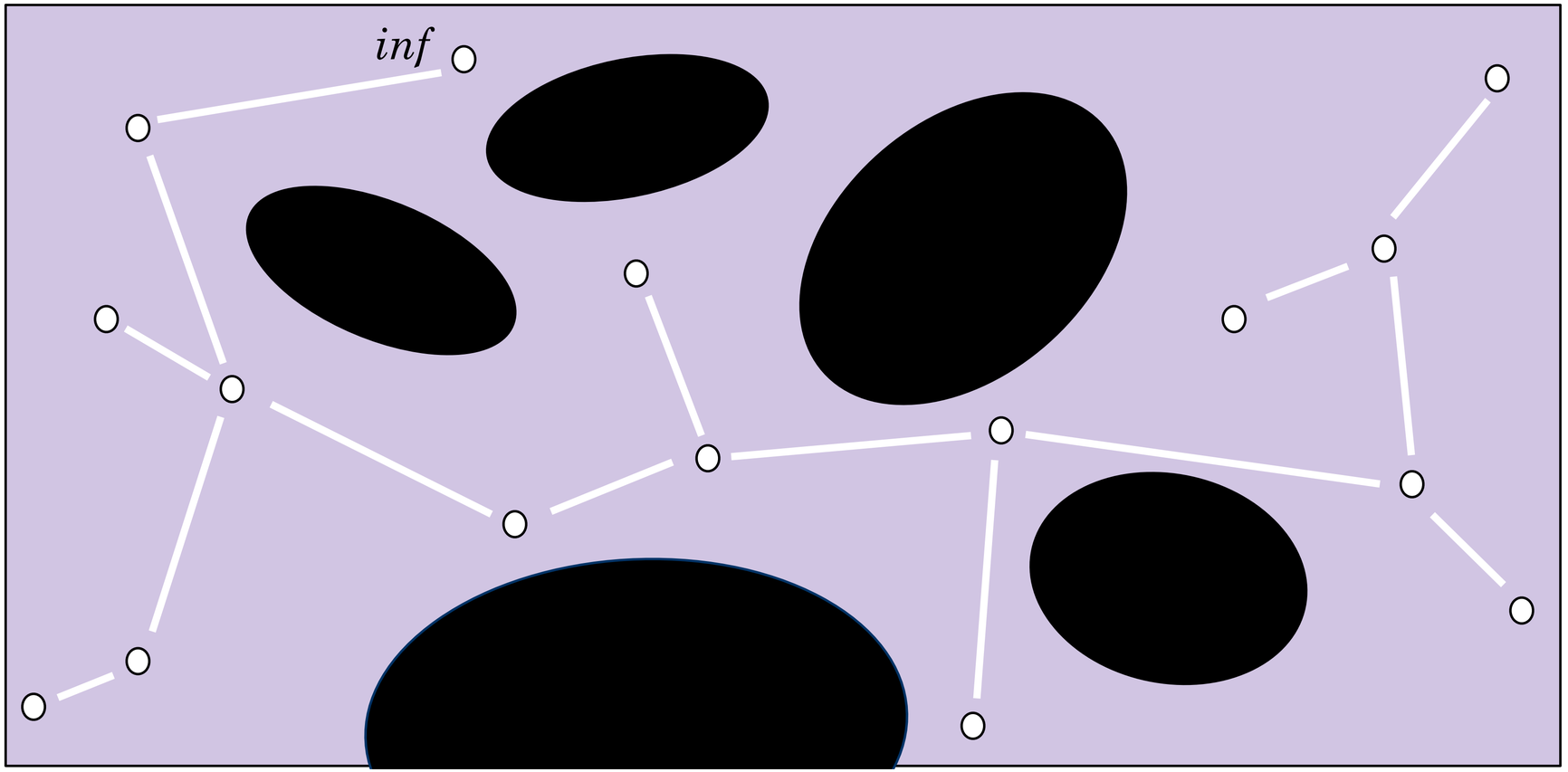

inf
1.00
1.00
0.50
1.00
0.50
0.33
0.50
0.33
inf
1.00
1.00
1.00
1.00
1.00

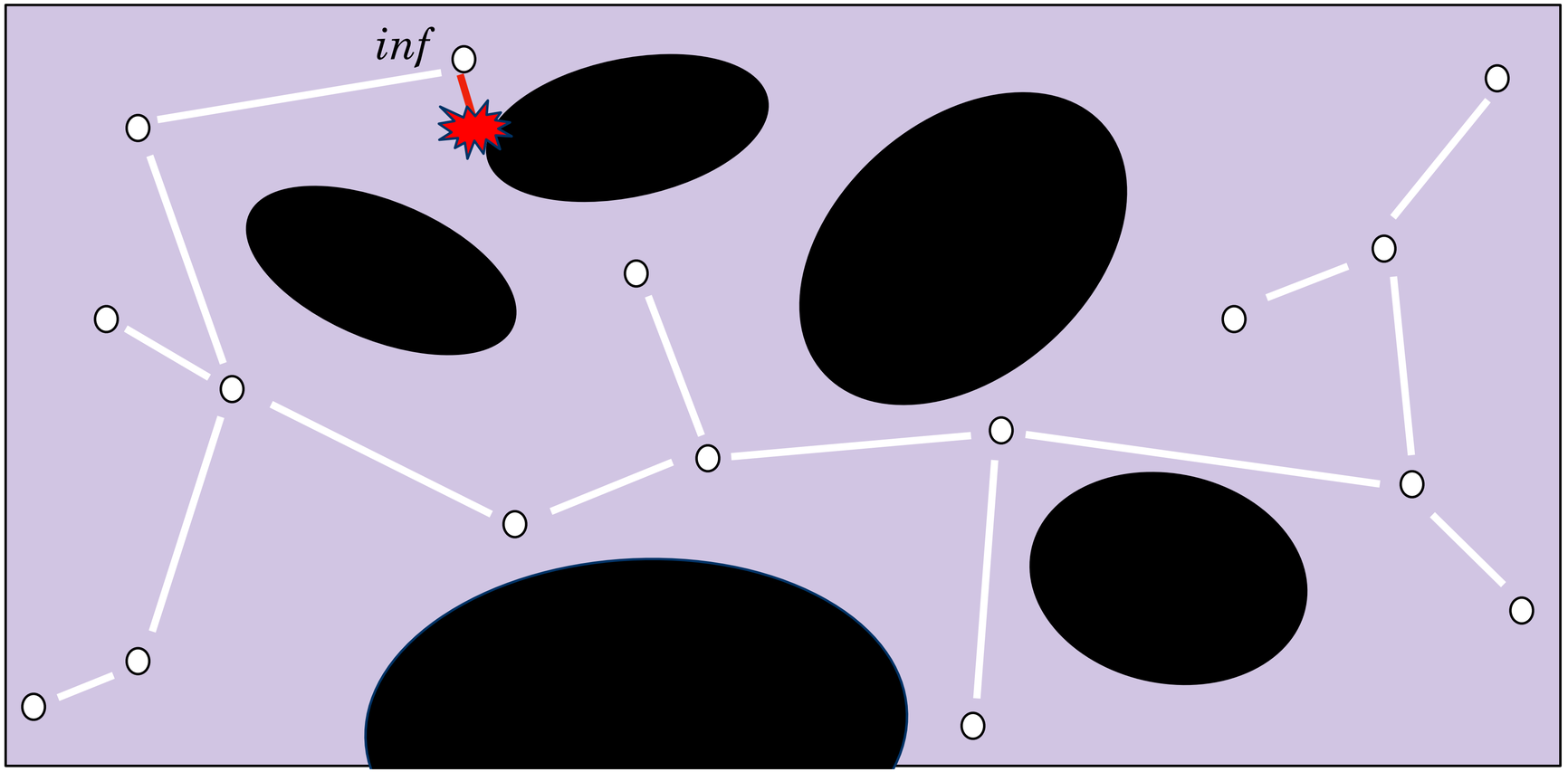
Learning Phase

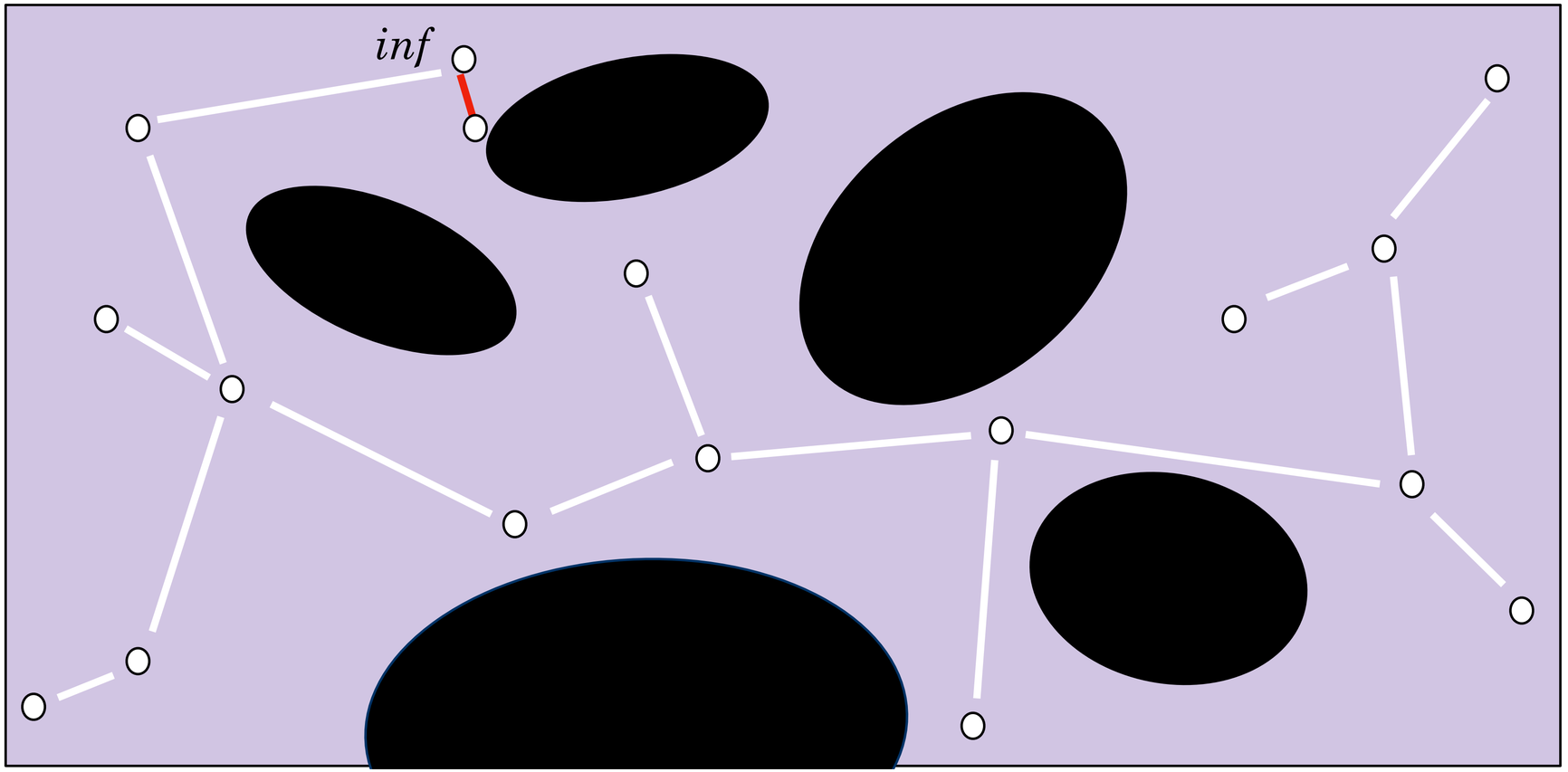
2. Expansion Step

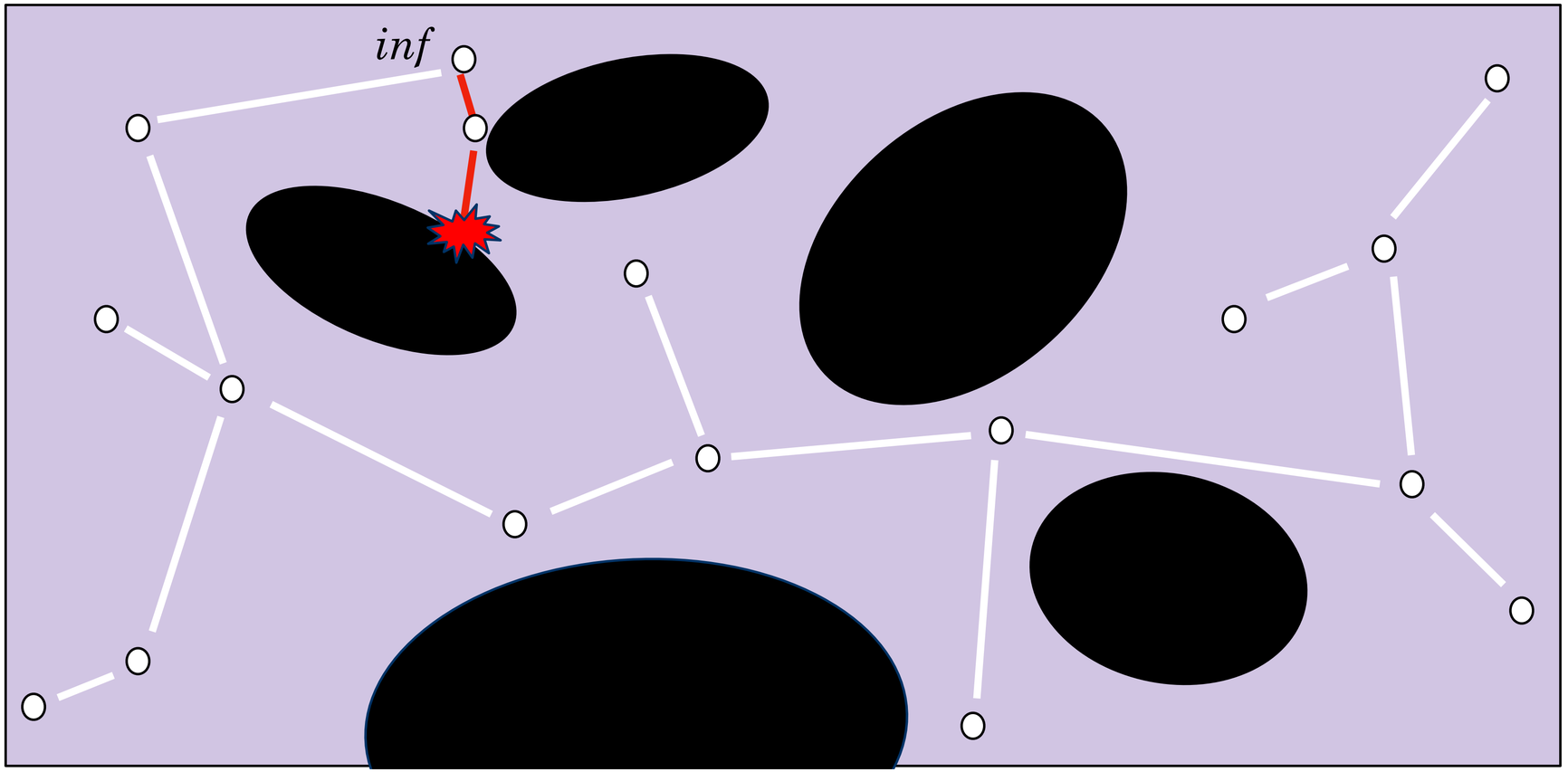
Loop

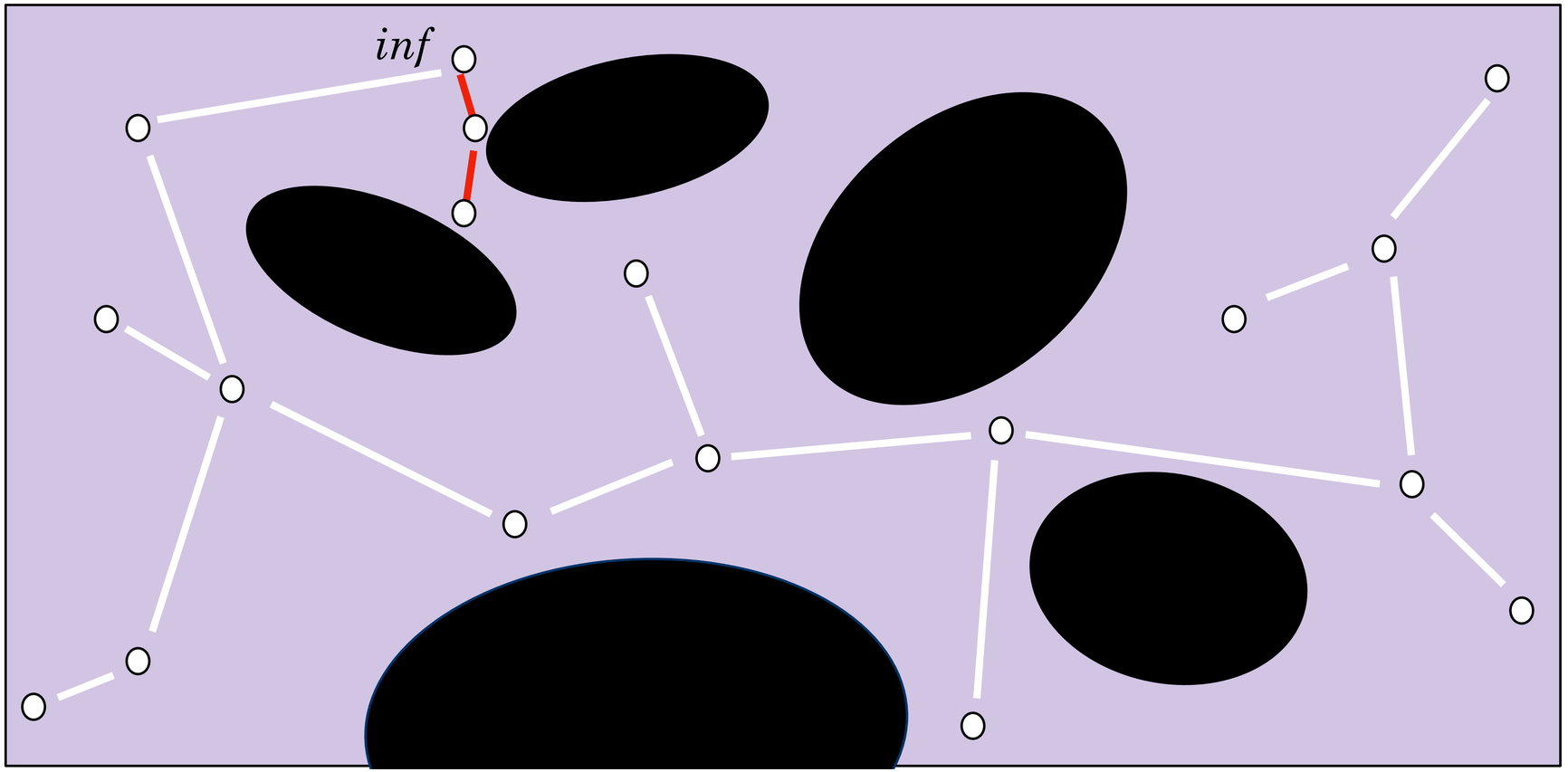
1. Pick a random direction of motion in *C*-space
2. Move in the direction until an obstacle is hit
3. Check for connection with another node
4. Repeat until the path can be connected to another node

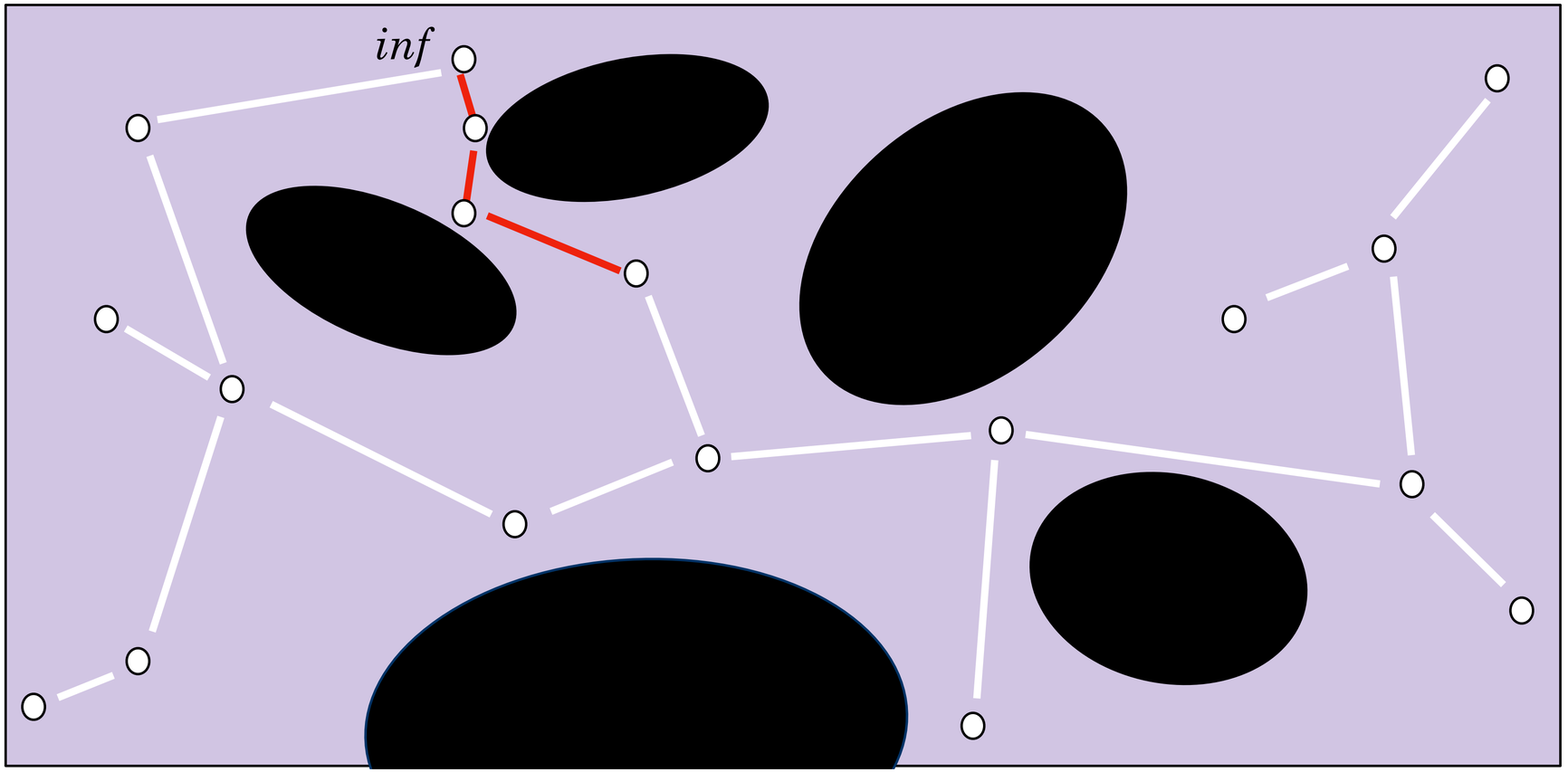

inf


inf


inf

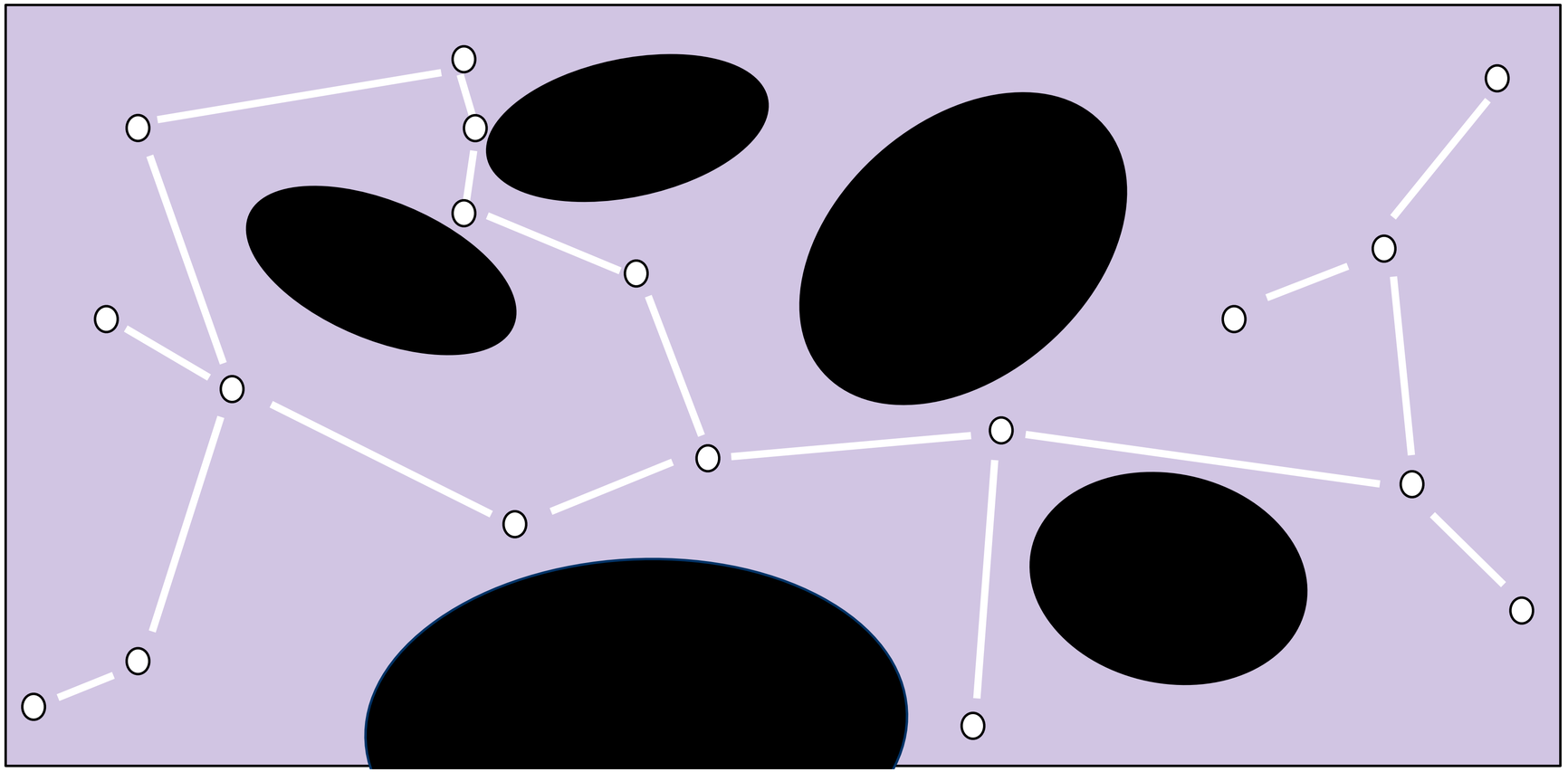

inf


inf


inf

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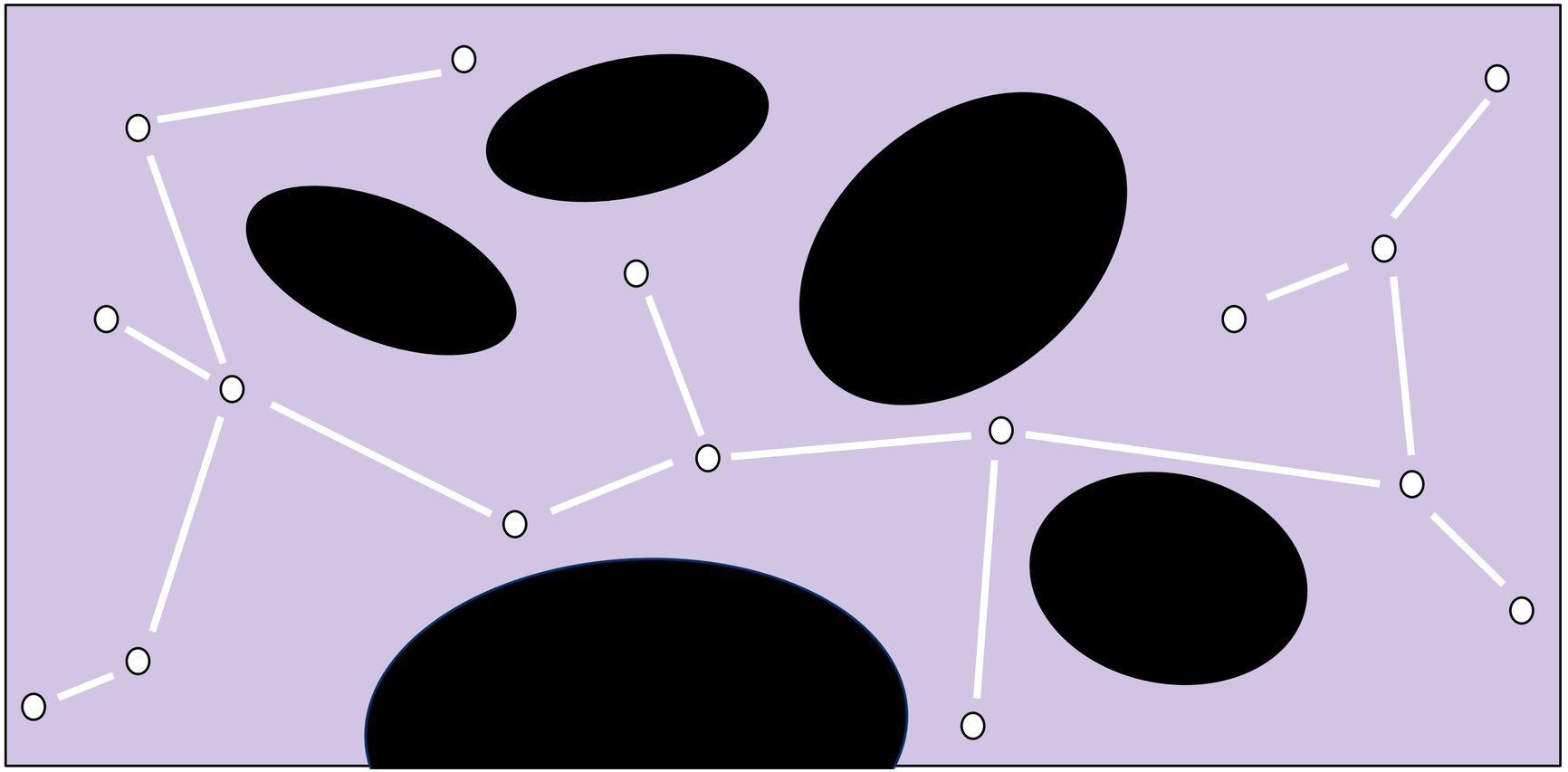


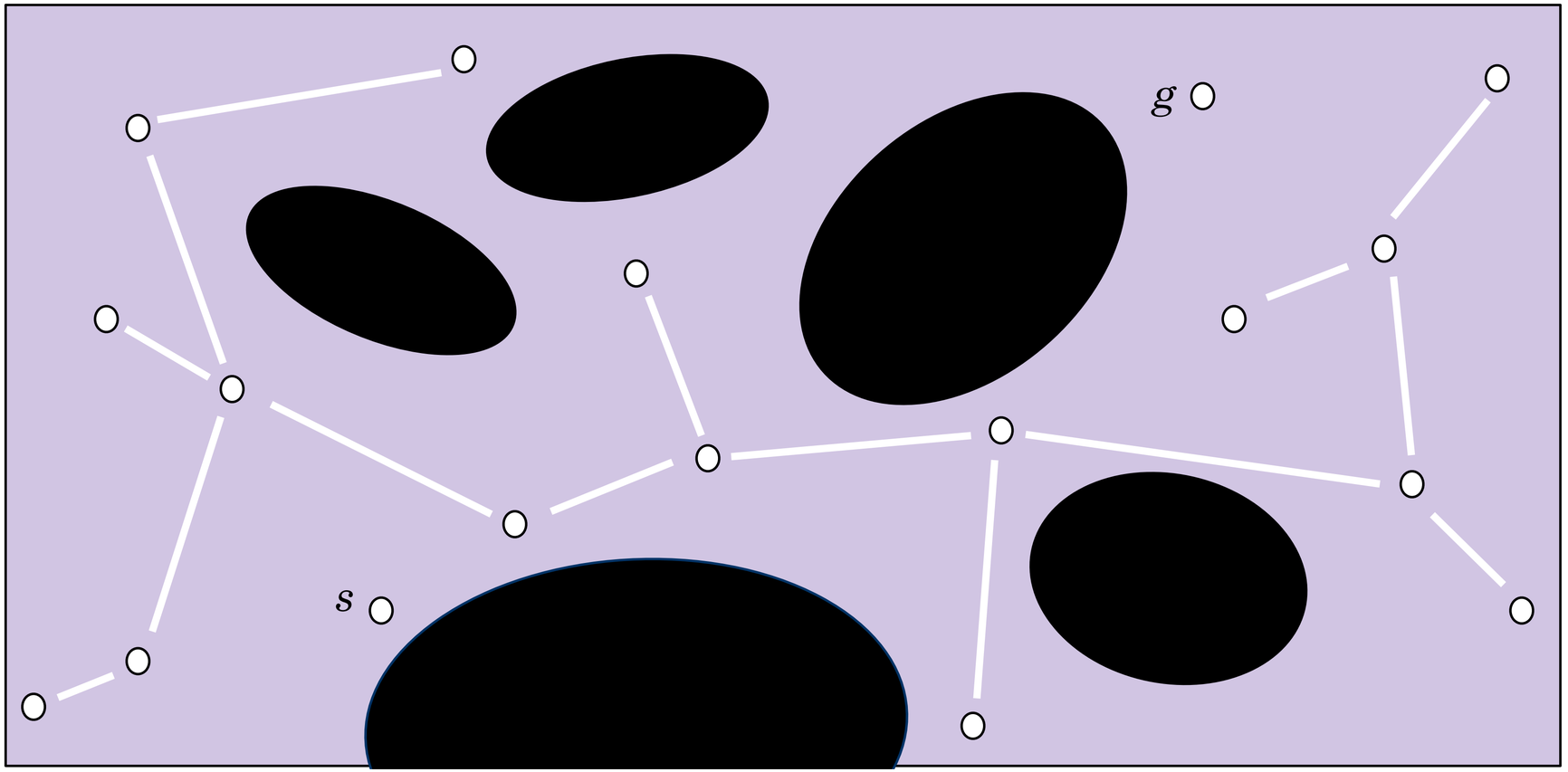
Query Phase

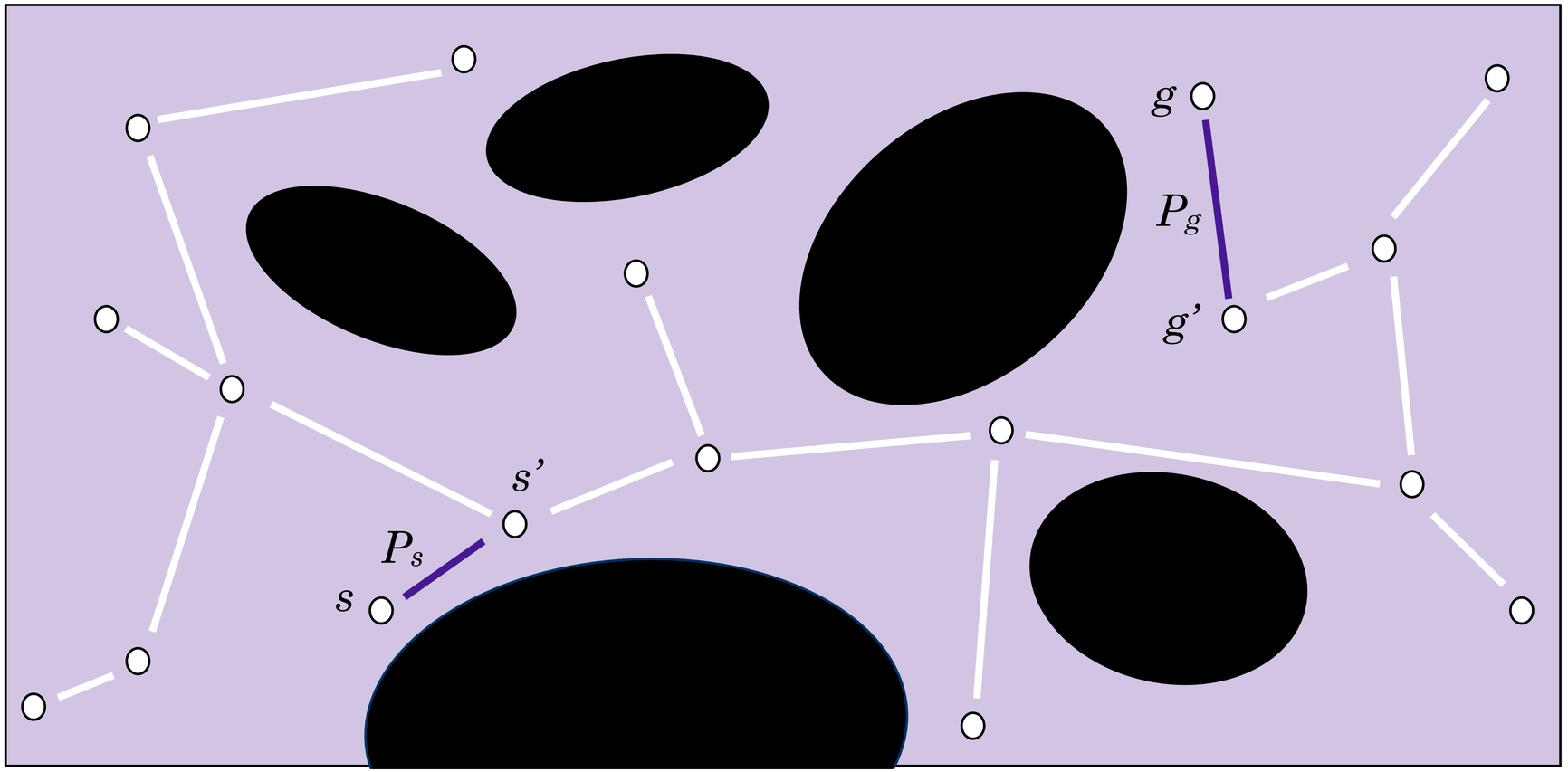
1. Given the start and goal configurations s and g, calculate feasible paths P_s and P_g to the nodes s' and g' on the roadmap (w/ LP)
2. Calculate the path P from s' to g' using the roadmap and a **tree search** planner

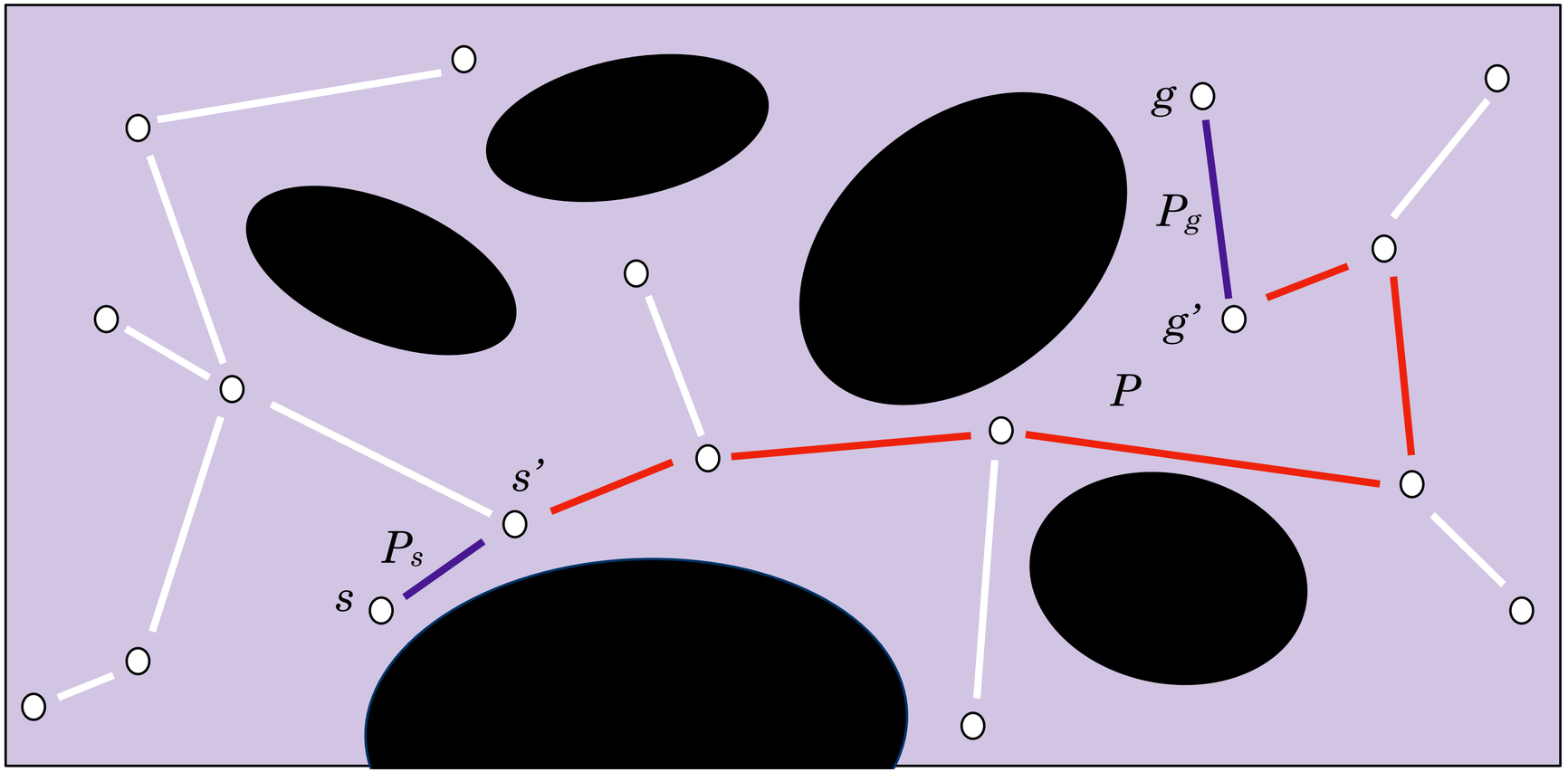
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g
s


g
P_g
g'
s'
P_s
s


g
P_g
g'
P
s'
P_s
s

Probablistic Road Maps

Two Tenets

1. **Efficient collision checking** of sampled configurations and connections between nodes
2. A **small number of milestones** and local paths are capture the connectivity of the free space.

Agenda

1. Algorithm run-through
2. Pseudo Code
3. **Example Implementation**



Courtesy of T. Bretl